# IRREGULARITY VERSUS GROWTH RATE IN DISK-STACKING PHYLLOTAXIS

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Abstract. In a disk-stacking model of phyllotaxis, slowly decreasing diameters can produce successive Fibonacci transitions. We identify a geometric obstruction to uniform guarantees based on separate upper bounds for irregularity and diameter decrease. For each hexagonal front written as a single-notch word $D^mU^n$ with $m, n \geq 2$ and $n/2 < m < 2n$, we construct an explicit open cone of perturbations for which the first placement is a pentagon transition. Both directional deviations tend to zero, all initial disks obey the diameter law, and the rate tends to zero with the perturbation. The proof compares all competing placements at a four-contact lattice site. We also derive uniform local transition formulas and a cancellation estimate: during a rhombic phase of $L$ placements, ratios within either vector family change by $O(|c| + c^2L)$. These results explain how inherited irregularity can compete with the $O(|c|)$ clearance created by a triangle transition. Simulations from three-disk fronts suggest a crossover controlled by the mismatch-to-rate ratio. A first-order event calculation supports delayed breakdown, but its transfer to the nonlinear dynamics remains open. Lean 4 verifies specified local identities and the cancellation estimate; the complete geometric argument is not formalised.

## 1. Introduction

Pairs of consecutive Fibonacci numbers, such as 34 and 55, are common in sunflower phyllotaxis, but exceptions are well documented: in a citizen-science survey of 657 heads, 565 of the 768 parastichy numbers in the most reliable subset were Fibonacci numbers, 67 more had a Fibonacci structure of a predefined type, and some heads had no assignable parastichy number at all [11]. A good model of phyllotaxis should therefore explain both the rule and its exceptions. This paper is about one exception mechanism in a geometric model that produces Fibonacci transitions, and about the conditions needed to preserve them.

The model goes back to Schwendener [9] and has been studied in detail by Douady and Couder [4], Atela, Golé and Hotton [2], Atela and Golé [3], Golé, Dumais and Douady [6], Golé and Douady [5] and Swinton [10]. Disks representing young organs are stacked on a cylinder, each new disk at the lowest position where it touches two disks already present and overlaps none. Slowly decreasing diameters can generate the sequence of parastichy numbers $(1, 2) \to (3, 2) \to (3, 5) \to (8, 5) \to \cdots$. Each step is a *Fibonacci transition*, made of triangle transitions on the vector family whose count is held fixed (Section 2 gives the definitions). Golé, Dumais and Douady stated the regime in which this should happen as a conjecture [6, p. 30]:

For each prescribed pair of parastichy numbers and integer $N > 0$, the conjecture asks for positive tolerances $\varepsilon_1, \varepsilon_2, \varepsilon_3$ such that every front with irregularity below $\varepsilon_1$, notch angles below $60^\circ + \varepsilon_2$, and diameter-decrease rate below $\varepsilon_3$ undergoes $N$ Fibonacci transitions. The tolerances may depend on the pair and on $N$, but the admissible fronts and rates satisfy these three bounds independently.

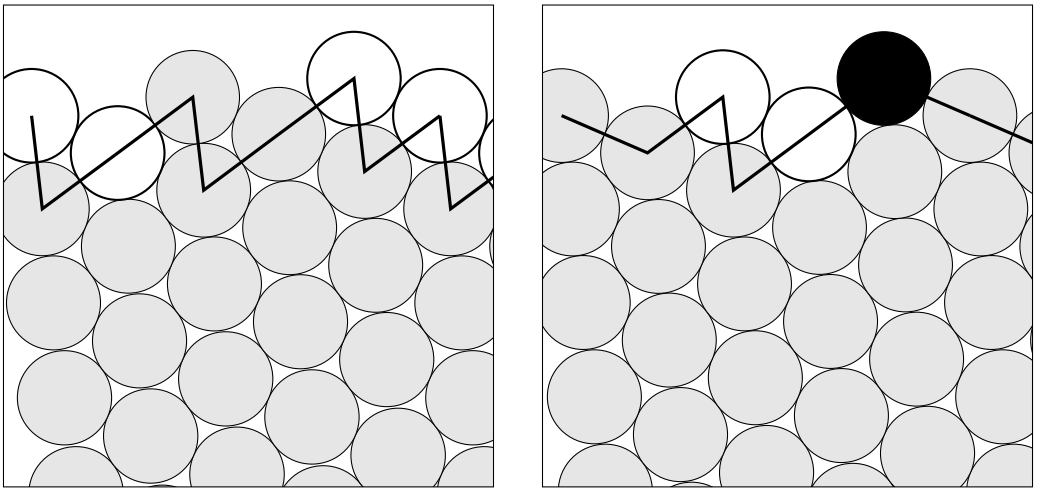
We test the *initial Fibonacci scenario*: the prescribed transitions must proceed from the given front without an intervening pentagon. Our first-disk construction rules out this guarantee. It does not rule out Fibonacci transitions occurring after an unrestricted transient; the word "undergoes" in the original statement should be read with this distinction in mind.

**An example.** Take a front of three disks: one down vector and two up vectors at angles $\beta \pm \varepsilon/2$ with $\beta = 0.523$ (just under $30^\circ$, notch angle $62.6^\circ$), and let the diameter decrease

*Date*: Preprint v2, 24 October 2026 (v1: 20 October 2026).
2020 *Mathematics Subject Classification.* 92C80, 37N25, 52C15.
*Key words and phrases.* Phyllotaxis, disk stacking, Fibonacci transitions, parastichy, formal verification.

| $\varepsilon$ | $c$ | $\varepsilon/\lvert c\rvert$ | First complete | Second-stage outcome | Third complete |
|---|---|---|---|---|---|
| 0.005 | $-0.0025$ | 2 | 367 | 1084: $(3,5)$ | 2196 |
| 0.005 | $-0.00125$ | 4 | 731 | 2163: P to $(3,3)$ | none |
| 0.01 | $-0.005$ | 2 | 185 | 544: $(3,5)$ | 1101 |
| 0.01 | $-0.0025$ | 4 | 369 | 1083: P to $(3,3)$ | none |

TABLE 1. Placement indices for four runs with $\beta = 0.523$ and diameter-consistent seeds. The first and third complete transitions reach $(3,2)$ and $(8,5)$, respectively; P denotes a pentagon. Equal ratios give the same qualitative outcome in these examples, while event indices scale approximately with $1/|c|$. Full histories accompany the code.



(a) $\varepsilon/|c| = 2$: three triangles, front $(3,5)$

(b) $\varepsilon/|c| = 4$: two triangles, then a pentagon

FIGURE 1. The top of the stack at the end of step two, for the first two runs of Table 1 ($\beta = 0.523$, $\varepsilon = 0.005$, $c = -0.0025$ and $-0.00125$), drawn from the simulator (`verify/pattern_figure.py`). The cylinder is unrolled, with circumference 1, and a little more than one turn is shown. Open disks are the triangle transitions of step two, the black disk is the pentagon, and the thick line is the front. In (a) the front ends with 5 up vectors; in (b) the pentagon removes one again.

linearly with height at rate $c$. With $\varepsilon = 0.005$ and $c = -0.0025$ the stack passes $(1,2) \to (3,2) \to (3,5) \to (8,5)$ cleanly. Halve the rate to $c = -0.00125$ and keep $\varepsilon$: after $(3,2)$ come two triangle transitions to $(3,3)$ and $(3,4)$, and at disk 2163 a pentagon transition returns the front to $(3,3)$. The slower run develops the defect. Halving $|c|$ doubles $\varepsilon/|c|$, from 2 to 4 (Table 1 and Figure 1).

**Results.** Our main result is an immediate obstruction. We then examine how the same local geometry can reappear after a long rhombic phase.

The immediate failure happens at hexagonal fronts. A rhombic lattice front whose notch hinge is exactly $60^\circ$ is a piece of the hexagonal packing, and it exists for every $(m,n)$ with $m,n \geq 2$ and $n/2 < m < 2n$ (Lemma 4.1); its up vectors have bearing $u$ with $\tan u = \sqrt{3}m/(2n-m)$. Write $H(\alpha,\sigma,\varepsilon,c)$ for the same front with three hinges changed by small amounts, $180^\circ + \alpha$ between the last two down vectors, $60^\circ + \sigma$ at the notch and $180^\circ - \varepsilon$ between the first two up vectors, the ridge taking up the difference, and with every disk, the initial ones included, of diameter (2) (Definition 4.2). Put $\kappa_u = \cos(u - 60^\circ)$.

**Theorem A** (Pentagon on the first disk)**.** *Let $m, n \geq 2$ with $n/2 < m < 2n$, and let $a, s, e$ and $q > 0$ satisfy*

$$\max\Big(e - a + \frac{q}{2}\kappa_u,\ -\frac{q}{2}\kappa_u\Big) \ < \ s \ < \ e - \frac{q}{4}\big(\sqrt{3}\sin u + 5\cos u\big). \tag{1}$$

*Then for all sufficiently small $t > 0$, $H(ta, ts, te, -tq)$ is a valid front and the first disk placed on it is a pentagon.*

The two lower bounds say only that no two disks of the front overlap, at the ridge and at the notch (Lemma 4.3). The upper bound is the real condition: the pentagon disk must clear its neighbour. All three hold, for example, when $e > 0$, $a = e/2$, $s = 3e/4$ and $0 < q \leq e/20$. For $(2, 3)$, $\kappa_u = 5/(2\sqrt{7})$ and the upper bound is $s < e - \frac{13\sqrt{7}}{28}q$. The rate enters with a sign: shrinking pushes the pentagon disk onto its neighbour, so a pentagon needs the rate to be small compared with the mismatch. Along such a ray both directional deviations are $O(t)$ and the product irregularity is $O(t^2)$; see (4). Thus the construction also applies when both vector families are required to be nearly regular.

**Corollary 1.1** (No independent tolerance box)**.** *Fix $m, n \geq 2$ with $n/2 < m < 2n$. For every $\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0$, there is a valid front with both directional deviations below $\varepsilon_1$, notch angle below $60^\circ + \varepsilon_2$, and rate below $\varepsilon_3$, whose first placement is a pentagon.*

*Proof.* Take $(a, s, e, q) = (\frac{1}{2}, \frac{3}{4}, 1, \frac{1}{20})$ in Theorem A. The two directional deviations tend to zero with $t$, the notch angle is $60^\circ + 3t/4$, and the rate is $t/20$. Choose $t > 0$ small enough to satisfy all three prescribed bounds and the conclusion of the theorem. □

**Relation to the constant-diameter classification.** Golé and Douady [5] work at constant diameter. They classify the offspring of every notch with at most three vectors by its hinge angles and one bearing [5, Sec. 4], and use this classification to prove that stacking from any cylindrical chain with parastichy numbers $(1, 2)$ or $(2, 1)$ converges to a rhombic tiling [5, Thm. 1]. Their reduction from a whole front to its short notches is proved for the chains that arise there [5, Props. 4.4, 4.5]; for longer chains they give simulations and heuristic arguments [5, Sec. 2.4]. Theorem A overlaps with their work in exactly one place. At $q = 0$ the disks $A, B, C, E$ form a notch with one down and two up vectors, with hinges $\tau = 60^\circ + \sigma$ and $\gamma = 180^\circ - \varepsilon$, and their pentagon condition $\gamma + \tau \leq 240^\circ$ reads $\sigma \leq \varepsilon$, the zero-rate form of the upper bound in (1). Three things are new.

(1) *The whole periodic front.* Near a hexagonal front, the lowest site is not decided inside one short notch. Six parent pairs meet at it, including $Z$ and $E$, which lie two diameters apart across a notch with two down and two up vectors, and the result has to hold for every $(m, n)$ with $m, n \geq 2$. Lemma 4.4 and the proof of Theorem A settle this degenerate site and exclude every other site of the front.

(2) *The rate.* Shrinking moves the pentagon boundary by $-\frac{q}{4}(\sqrt{3}\sin u + 5\cos u)$, so it works against the pentagon. It also closes the ridge, so the hexagonal front is no longer valid once $c < 0$; every valid front near it carries a second irregularity (Lemma 4.3). The window compares the mismatch with the rate on the same scale. It is a first-disk counterpart of the ratio law described below.

(3) *The conjecture.* The tolerance conjecture of [6] requires $c < 0$, and it is not treated in [5]. Corollary 1.1 alone could also be obtained from the case $q = 0$ of the whole-front argument, by perturbing $c$ at fixed $t$; item 1 is needed either way. The first-order rate term adds how small the rate must be: below a fixed multiple of the mismatch.

This obstruction uses fronts whose notch excess is smaller than their directional mismatch. Requiring the mismatch to be small relative to the initial notch excess excludes this particular construction. The three-disk family below tests whether that protection survives later transitions: each slow triangle leaves a notch excess on the much smaller scale $|c|$, while the cancellation estimate allows an initial mismatch to persist.

**The delayed mechanism.** For this we study a three-disk family. Write $F(\beta, \varepsilon)$ for the $(1, 2)$ front of the example (Definition 2.1), with $\beta$ in the interval $(\beta_{\min}, \pi/6)$, $\beta_{\min} = \arctan(-\sqrt{3}/2) + \pi/3 \approx 19.107°$, on which it is a valid front that has not yet reached a triangle transition. Its notch angle exceeds 60° by an amount fixed by $\beta$, far more than its irregularity, so Theorem A does not apply to it.

**Theorem B** (First transition)**.** *For every $\beta \in (\beta_{\min}, \pi/6)$ there are $\varepsilon_0, c_0 > 0$ such that for $|\varepsilon| < \varepsilon_0$ and $-c_0 < c < 0$, stacking from $F(\beta, \varepsilon)$ performs two triangle transitions, on different up vectors, and no pentagon transition before its front first has parastichy numbers $(3, 2)$.*

Theorem B is the first step of the delayed mechanism; its main tool is the cancellation estimate of Proposition 5.2, which carries the mismatch $\varepsilon$ through long rhombic phases with an error of order $|c|$. Whether the inherited mismatch later produces a pentagon, between $(3, 2)$ and $(8, 5)$, is not proved here. An exact first-order event calculation and the simulations support it, and we state it as Conjecture C in Section 7, together with what a proof still needs. Nothing in Theorems A and B or Corollary 1.1 depends on it.

The numerics in Section 8 locate the threshold for $F(\beta, \varepsilon)$. Over thirty starting angles $\beta \in [0.40, 0.52]$ and three rates, none of the sampled second transitions breaks at sampled ratios $\varepsilon/|c| \le 2.3$, and all break at sampled ratios $\varepsilon/|c| \ge 3.4$ (Figure 11). We refer to this empirical crossover as the *ratio law*; it is not a monotonicity theorem or a universal threshold. Golé, Dumais and Douady observed that irregularity alone drives the pattern towards quasi-symmetry [6, p. 29, item (iii)], and Swinton found that noise and rate "interact in a complex way" [10, Sec. 5.3] and noted that there is no theoretical guidance on the critical rate [10, Sec. 6]. The ratio law is a quantitative form of that interaction for one initial irregularity.

**The geometric mechanism.** Figure 8 shows the relevant local geometry. A triangle transition at a slow rate is nearly equilateral, so it leaves a notch whose hinge is 60° plus a cushion of order $|c|$ (Lemma 3.3). During the long rhombic phases between transitions the shape of the front is frozen to first order (Proposition 5.2), so an initial mismatch $\varepsilon$ between the up vectors is retained up to $O(|c|)$ over a phase of length $O(1/|c|)$. When two mismatched up vectors meet a 60° notch, a pentagon fits once the mismatch beats the cushion (Lemma 7.2). Slower growth reduces the local clearance scale. Establishing that this necessarily selects a pentagon later in the three-disk dynamics is the remaining issue in Conjecture C.

**Plan.** Section 2 sets up the model. Section 3 gives the first-order rhombus and triangle steps, and Section 4 uses them to prove Theorem A. Section 5 proves that front shape is frozen through rhombic phases. Section 6 proves Theorem B; Section 7 develops the evidence for Conjecture C. Section 8 reports the ratio law and Section 9 discusses a narrower conjecture and open problems.

## 2. The model

2.1. **Disks, fronts and transitions.** Work on the cylinder $\mathbb{R}/\Lambda\mathbb{Z} \times \mathbb{R}$, identified with a strip of the complex plane, with height $y = \operatorname{Im} z$. A disk centred at $z$ has diameter

$$d(z) = b_0 + c \operatorname{Im} z, \qquad c < 0, \tag{2}$$

so disks placed higher are smaller, and $|c|$ is the rate of decrease per unit of height (the rate of [6]). Two disks touch when the distance between their centres is the mean of their diameters. Each new disk is placed at the lowest admissible position above the front where it touches two front disks and overlaps no disk, including periodic images [6, 5]. Only positions with positive diameter are considered. The transition statements concern this upper placement branch; contacts below the front do not define placements.

A *front* is a closed chain of disks $z_0, z_1, \ldots, z_{N-1}$ winding once around the cylinder, consecutive disks touching, on top of the stack [3, 5]. Its *front vectors* are $v_k = z_{k+1} - z_k$, read from left to right. A vector pointing down ($\operatorname{Im} v_k < 0$) is a *down vector*; the others are *up vectors*, following the convention that a horizontal vector is up [6]. We write $(m, n)$ for the numbers of

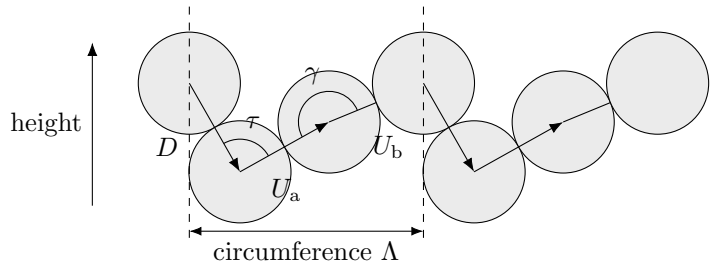
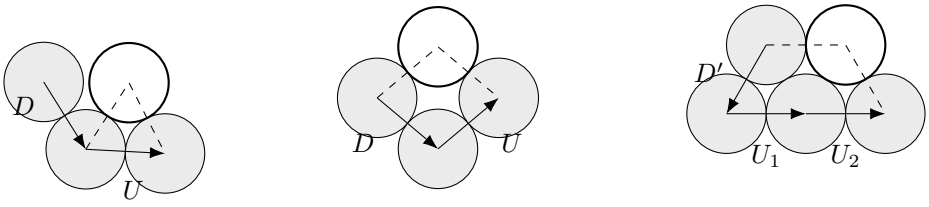



FIGURE 2. The front $F(\beta, \varepsilon)$ of Definition 2.1, two turns of the unrolled cylinder, with the up-vector mismatch $\varepsilon$ exaggerated. It has one down vector $D$ and two up vectors, so its parastichy numbers are $(1, 2)$. The notch hinge $\tau$ sits between $D$ and $U_{\mathrm{a}}$; the hinge $\gamma$ between the two up vectors differs from $180°$ by $\varepsilon$.




T: hinge $> 120°$ R: hinge in $(60°, 120°)$ P: $\gamma + \tau \le 240°$

FIGURE 3. The three transitions. Grey disks belong to the front, the open disk is new, and dashed segments join it to the two disks it touches. The pentagon panel is drawn at the degenerate point $\gamma + \tau = 240°$, where the new disk touches all three of $P_0$, $P_2$, $P_3$ (Lemma 7.2).

down and up vectors; these are the parastichy numbers of the front. The *hinge* at $z_{k+1}$ is the upper angle between the two vectors,

$$h(v_k, v_{k+1}) = \arg v_k + \pi - \arg v_{k+1}, \tag{3}$$

read as: the angle above the chain at the joint. A *notch* is a down vector followed by an up vector. Its hinge is the notch angle $\eta$ of [6]; in bearing angles, $\eta = \pi - (\arg U - \arg D)$. Figure 2 shows the smallest example, the front with one down and two up vectors used throughout this paper.

For the up and down vector families, define their complex means $\mu_U$, $\mu_D$ and deviations

$$\eta_U = \frac{\max_j |U_j - \mu_U|}{|\mu_U|}, \qquad \eta_D = \frac{\max_k |D_k - \mu_D|}{|\mu_D|}, \qquad I = \eta_U \eta_D, \qquad J = \max(\eta_U, \eta_D). \tag{4}$$

The means are nonzero for the fronts considered here. The irregularity of [6] is the product $I$. We also record $J$, which controls both directional deviations separately. For a single down vector $\eta_D = 0$, so $I = 0$ even when the up vectors differ. The first-disk construction has $J \to 0$ as well as $I \to 0$ and does not rely on this degeneracy.

A new disk touching $z_a$ and $z_b$ replaces the disks strictly between them in the front. There are three cases (Figure 3):

- *triangle* (T): $b = a + 1$. One vector is replaced by two; the front gains a vector.
- *rhombus* (R): $b = a + 2$. A notch $D, U$ is replaced by $U', D'$; the vector count is kept.
- *pentagon* (P): $b = a + 3$. Three vectors are replaced by two; the front loses a vector.

At constant diameter, Golé and Douady classified the offspring of notches by their hinge angles. A notch with one down and one up vector gives a rhombus when its hinge lies in $[\pi/3, 2\pi/3]$ and a triangle, resting on the less steep of its two vectors, when the hinge lies in $[2\pi/3, \pi]$. A notch $D, U_1, U_2$ can give a pentagon only when its two hinges $\tau = h(D, U_1)$ and $\gamma = h(U_1, U_2)$ satisfy $\gamma + \tau \le 4\pi/3$ [5, Prop. 4.3].

2.2. **The initial fronts.**

**Definition 2.1.** For $\beta \in (\beta_{\min}, \pi/6)$ and small $\varepsilon$, the front $F(\beta, \varepsilon)$ consists of three touching disks whose diameters follow (2), with front vectors $D, U_{\mathrm{a}}, U_{\mathrm{b}}$, where $\arg U_{\mathrm{a}} = \beta + \varepsilon/2$, $\arg U_{\mathrm{b}} = \beta - \varepsilon/2$ and $D$ is the down vector closing the chain: $\mathrm{Im}(D + U_{\mathrm{a}} + U_{\mathrm{b}}) = 0$. The circumference is $\Lambda = \mathrm{Re}(D + U_{\mathrm{a}} + U_{\mathrm{b}})$.

At $c = 0$ all three disks have diameter $b_0$. One could instead give the initial disks a common diameter, treating the first front as data outside the law (2); that changes the front by $O(|c|)$. The analytic first-transition argument is stable under this change, but it shifts the measured threshold of the ratio law (Section 8), because a defect of order $|c|$ is itself an irregularity on the scale that matters.

So $U_{\mathrm{a}}$ is the steeper up vector when $\varepsilon > 0$. For $\varepsilon = c = 0$ this is the $(1, 2)$ rhombic lattice with $\sin \beta_D = -2 \sin \beta$, and its notch angle is $\pi - \beta + \beta_D$, which decreases to $60°$ as $\beta \to \pi/6$ (it is $62.6°$ at $\beta = 0.523$). At $\beta = \beta_{\min}$ the notch angle is $120°$ and the front is already at its triangle threshold (Lemma 2.2).

A *Fibonacci transition* from a front with $(m, n)$, $n$ up vectors, is a sequence of $n$ triangle transitions, one on each up vector, with rhombic phases in between and no pentagon, ending at $(m + n, n)$; the next one exchanges the roles of up and down vectors. This is the scenario of [6, p. 25]. From $F(\beta, \varepsilon)$ the first Fibonacci transition is $(1, 2) \to (2, 2) \to (3, 2)$ and the second is $(3, 2) \to (3, 3) \to (3, 4) \to (3, 5)$.

2.3. **Limiting tilings.** Three configurations at constant diameter organise the proofs, because the slow dynamics stays close to them: the $(1, 2)$ lattice at its triangle threshold, the $(2, 2)$ front it leaves after one triangle, and the tiling $K_0$ reached from the $(3, 2)$ lattice at its threshold after two triangles of step two. At a threshold the up vectors are exactly $60°$ above the down vectors, so that a notch hinge equals $120°$.

**Lemma 2.2** (Threshold law)**.** *A rhombic lattice front with $m$ down vectors of angle $\beta_D$ and $n$ up vectors of angle $\beta_D + \pi/3$ closes, that is $m \sin \beta_D + n \sin(\beta_D + \pi/3) = 0$, if and only if*

$$\tan \beta_D = -\frac{\sqrt{3}\, n}{2m + n}.$$

*For $(m, n) = (1, 2)$ this gives $\tan \beta_D = -\sqrt{3}/2$, $\beta_D \approx -40.893°$, up angle $\beta_{\min} \approx 19.107°$. For $(m, n) = (3, 2)$ it gives $\tan \beta_D = -\sqrt{3}/4$, $\beta_D \approx -23.413°$, up angle $36.587°$.*

In words: the more down vectors per up vector, the flatter the down vectors are when the first triangle becomes possible.

*Proof.* Expand $\sin(\beta_D + \pi/3) = \frac{1}{2} \sin \beta_D + \frac{\sqrt{3}}{2} \cos \beta_D$: the closing condition becomes $(2m + n) \sin \beta_D + \sqrt{3}\, n \cos \beta_D = 0$. □

Both thresholds are independent of $\beta$ and $\varepsilon$. This is why the constants below are universal: the starting angle only sets how long the slow drift takes to reach them. A convenient exact model: in units of the diameter, the $(3, 2)$ threshold vectors are

$$D = \tfrac{1}{\sqrt{19}}(4, -\sqrt{3}), \quad U = \tfrac{1}{2\sqrt{19}}(7, 3\sqrt{3}), \quad S = \tfrac{1}{2\sqrt{19}}(1, -5\sqrt{3}), \tag{5}$$

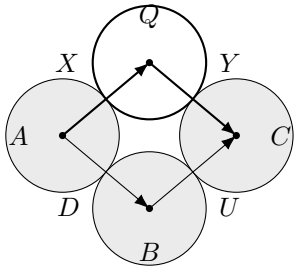
with $U$ and $S$ the rotations of $D$ by $+60°$ and $-60°$. Writing points as $(X/\sqrt{19}, \sqrt{3}\, Y/\sqrt{19})$ makes every point of these tilings rational in $(X, Y)$, and squared distances equal $(\Delta X^2 + 3\Delta Y^2)/19$. All distances quoted for the tiling $K_0$ below are exact rational identities in this model; the $(1, 2)$ threshold works the same way with 7 in place of 19. These identities are checked in Lean (`Tilings.lean`).

## 3. First-order transitions

Rhombic phases last about $1/|c|$ steps, so a uniform error of order $c^2$ per step contributes order $|c|$ over such a phase. This makes uniform local estimates essential. This section computes them.

| | |
|---|---|
| $c<0$, $b_0$ | rate of decrease of the diameter, diameter at height 0 |
| $\beta$, $\varepsilon$ | mean angle and mismatch of the two up vectors of $F(\beta,\varepsilon)$ |
| $D$, $U$; $\beta_D$, $\beta_U$ | down and up vectors and their bearing angles |
| $U_{\mathrm{a}}$, $U_{\mathrm{b}}$; $U^+$, $U^-$ | the two up vectors; the up and down vectors a triangle on $U$ creates |
| $h$, $\tau$, $\gamma$ | hinge (3); notch hinge; hinge between two up vectors |
| $\omega$ | $e^{i\pi/3}$, rotation by 60° |
| $\sigma$, $\delta$ | cushion above 60° after a triangle; overshoot of a hinge past 120° |
| $\theta$ | mismatch between the two up vectors of a pentagon notch |
| $\kappa_1$, $\kappa_2$ | loss of mismatch through step one; pentagon correction (Lemma 7.2) |
| $K_0$, $K^*$ | limiting tiling at the pentagon (Lemma 7.3); measured threshold of $\varepsilon/\lvert c\rvert$ |

TABLE 2. Notation.




$$X = U\big(1 + ic\,e^{i\beta_D}\big) + O(c^2)$$
$$Y = D\big(1 - ic\,e^{i\beta_U}\big) + O(c^2)$$

FIGURE 4. A rhombus step. At $c = 0$ the new cell is the rhombus $X = U$, $Y = D$; with shrinking disks each new vector is the old one turned and stretched by a factor that depends only on the other vector of the notch (Lemma 3.2).

3.1. **The apex branch.** Fix a notch, let the rate $c$ vary, and keep the disks of the notch touching their neighbours. The new disk is a transversal intersection of two tangency conditions, so it moves smoothly with $c$.

**Lemma 3.1** (Apex branch)**.** *Let $A(c)$ and $C(c)$ be $C^1$ families of centres and let $q_0$ satisfy $|q_0 - A(0)| = |q_0 - C(0)| = b_0$ with $q_0 - A(0)$ and $q_0 - C(0)$ not parallel. Then there is a unique $C^1$ family $Q(c)$ near $c = 0$ with $Q(0) = q_0$ such that the disk at $Q(c)$ touches the disks at $A(c)$ and $C(c)$ under* (2)*.*

*Proof.* The two tangency conditions $|Q - A|^2 = (b_0 + c(\operatorname{Im} A + \operatorname{Im} Q)/2)^2$ and the same with $C$ define a $C^1$ map of $(c, Q)$ whose partial derivative in $Q$ at $(0, q_0)$ is $v \mapsto \big(2(q_0-A)\cdot v,\ 2(q_0-C)\cdot v\big)$, which is invertible exactly when the two contact normals are not parallel. The implicit function theorem gives $Q$. □

Differentiating one tangency condition at $c = 0$, where all diameters equal $b_0$, gives the identity used in every proof below:

$$(Q - P)\cdot(Q' - P') = \tfrac{b_0}{2}\,(\operatorname{Im} P + \operatorname{Im} Q). \tag{6}$$

In words: the rate at which the gap between two touching disks opens equals the rate at which their mean diameter grows, which at $c = 0$ is half the sum of their heights times $b_0$.

3.2. **The rhombus step.**

**Lemma 3.2** (Rhombus step)**.** *Let $A \xrightarrow{D} B \xrightarrow{U} C$ be a notch of touching disks with $D$ and $U$ not parallel, and let $Q$ be the apex branch touching $A$ and $C$ with $Q = A + U$ at $c = 0$. Put $X = Q - A$, $Y = C - Q$ and $\xi = X - U$. Then*

$$\xi'(0) = \frac{i\,U D}{b_0} \quad \textit{(complex product)},$$

*so that, with $D = b_0 e^{i\beta_D}$ and $U = b_0 e^{i\beta_U}$,*

$$X = U\big(1 + ic\,e^{i\beta_D}\big) + O(c^2), \qquad Y = D\big(1 - ic\,e^{i\beta_U}\big) + O(c^2).$$

The formula says that the up vector is multiplied by a factor that depends only on the down vector it passes, and the down vector by a factor that depends only on the up vector. Proposition 5.2 is built on that.

*Proof.* Apply (6) to the four touching pairs $AB$, $BC$, $AQ$, $QC$. At $c = 0$, $Q - A = U$ and $C - Q = D$, and the heights satisfy $\operatorname{Im} Q = \operatorname{Im} A + \operatorname{Im} U$, $\operatorname{Im} C = \operatorname{Im} B + \operatorname{Im} U$. Subtracting the equations for $BC$ from $AQ$, and $QC$ from $AB$, the unknown motions of $A, B, C$ cancel and

$$U \cdot \xi' = -b_0 \operatorname{Im} D, \qquad D \cdot \xi' = -b_0 \operatorname{Im} U.$$

This $2 \times 2$ system has a unique solution because $U$ and $D$ are not parallel, and $\xi' = iUD/b_0$ satisfies it: with $|U| = |D| = b_0$, $U \cdot (iUD)/b_0 = \operatorname{Re}(\bar{U}\, iUD)/b_0 = -b_0 \operatorname{Im} D$, and similarly for $D$. □

3.3. **The triangle step and the cushion.**

**Lemma 3.3** (Triangle step)**.** *Let $B \xrightarrow{U} C$ be touching disks and $Q$ the apex branch touching both with $Q = B + \omega U$ at $c = 0$, where $\omega = e^{i\pi/3}$. Put $X = Q - B$, $Y = C - Q$ and $\xi = X - \omega U$. Then*

$$\xi'(0) = \frac{i\,U^2}{2b_0}, \qquad \text{so} \qquad X = \omega U + \tfrac{ic}{2} e^{i\beta_U} U + O(c^2), \quad Y = \bar{\omega} U - \tfrac{ic}{2} e^{i\beta_U} U + O(c^2).$$

*Consequently, to first order in $c$:*

(a) $\arg X - \arg(\omega U) = \frac{c}{2}\cos(\beta_U - \pi/3)$*;*
(b) $\arg X - \arg Y = \frac{2\pi}{3} + \frac{c}{2}\cos\beta_U$*.*

*If the triangle fires on the up vector of a notch whose hinge is $120^\circ + \delta$, the new notch at $B$ has hinge*

$$60^\circ + \delta + \tfrac{|c|}{2}\cos(\beta_U - 60^\circ) + O(c^2). \tag{7}$$

*Proof.* The three tangencies $BC$, $BQ$, $QC$ and (6) give, after the same cancellation as before, $(\omega U)\cdot\xi' = \frac{b_0}{2}\operatorname{Im}(\omega U - U)$ and $(\bar{\omega} U)\cdot\xi' = -\frac{b_0}{2}\operatorname{Im}(\omega U)$, whose unique solution is $iU^2/(2b_0)$. For (a), $X/(\omega U) = 1 + \xi/(\omega U)$ and the derivative of its argument is $\operatorname{Im}\big(iU/(2b_0\omega)\big) = \frac{1}{2}\cos(\beta_U - \pi/3)$; (b) adds the same computation for $Y/(\bar{\omega}U)$, using $\cos(\beta - \pi/3) + \cos(\beta + \pi/3) = \cos\beta$. The new hinge at $B$ is $h(D, X) = h(D, U) - (\arg X - \arg U) = 120^\circ + \delta - 60^\circ - \frac{c}{2}\cos(\beta_U - 60^\circ)$, and $c < 0$. □

At $c = 0$ and $\delta = 0$ the new hinge is exactly $60^\circ$: the new disk touches the disk below $B$ as well. That triple contact is the pentagon boundary, and (7) measures how far a slow triangle stays from it. The cushion is of order $|c|$; the overshoot $\delta$ is also of order $|c|$, since hinges drift by order $|c|$ per step (Corollary 3.4).

**Corollary 3.4** (Rotation rates)**.** *In the rhombus step of Lemma 3.2, to first order in $c$,* $\arg X - \arg U = c\cos\beta_D$ *and* $\arg Y - \arg D = -c\cos\beta_U$*.*

*Proof.* $X/U = 1 + \xi/U$ and $\operatorname{Im}(\xi'/U) = \operatorname{Im}(iD/b_0) = \cos\beta_D$; likewise $Y/D = 1 - \xi/D$. □

3.4. **Uniform remainders.** The derivative identities above hold for $C^1$ input families. A rhombic phase applies them about $1/|c|$ times, so it needs an $O(c^2)$ remainder that is uniform over all notches met along the way; $C^1$ regularity alone does not give that. Two observations supply it.

First, the law has no preferred scale. Translate a disk $B$ of diameter $b$ to the origin and divide all lengths by $b$. The law becomes $d = 1 + cy$ with the same dimensionless slope $c$, since $c$ is a ratio of a change in diameter to a change in height. Second, edge lengths are explicit. An edge of bearing $\beta$ leaving a disk of diameter $b$ ends at a disk of diameter $b + c\ell\sin\beta$, so tangency, $\ell = b + \frac{c}{2}\ell\sin\beta$, gives

$$\ell = \frac{b}{1 - \frac{c}{2}\sin\beta}. \tag{8}$$

**Lemma 3.5** (Uniform remainders)**.** *Let $\mathcal{K}$ be a compact set of bearing pairs $(\beta_D, \beta_U)$ with $\beta_U - \beta_D \notin \pi\mathbb{Z}$. There are $c_1 > 0$ and $R$ such that for every notch $A \xrightarrow{D} B \xrightarrow{U} C$ of touching disks with bearings in $\mathcal{K}$, whatever the diameter $b$ of $B$, and every $|c| \le c_1$, the rhombus apex exists and*

$$\frac{X}{U} = 1 + ic\,e^{i\beta_D} + r_X, \qquad \frac{Y}{D} = 1 - ic\,e^{i\beta_U} + r_Y, \qquad |r_X|, |r_Y| \le Rc^2.$$

*The same holds for the triangle step of Lemma 3.3 over any compact set of bearings $\beta_U$, and for any apex whose two contact normals at $c = 0$ are not parallel.*

*Proof.* By scale invariance take $B = 0$ and $b = 1$. By (8),

$$A = -\frac{e^{i\beta_D}}{1 + \frac{c}{2}\sin\beta_D}, \qquad C = \frac{e^{i\beta_U}}{1 - \frac{c}{2}\sin\beta_U},$$

which are real-analytic in $p = (\beta_D, \beta_U, c)$ for $|c| < 2$. The apex $Q$ solves $\Phi(Q, p) = 0$, where

$$\Phi(Q,p) = \Big(|Q-A|^2 - \tfrac{1}{4}(d_A + d_Q)^2,\ |Q-C|^2 - \tfrac{1}{4}(d_C + d_Q)^2\Big), \qquad d_Q = 1 + c\,\mathrm{Im}\,Q,$$

is polynomial in $Q$, $A$, $C$ and $c$, hence real-analytic. At $c = 0$ the solution is $Q_*(\beta) = A + e^{i\beta_U}$, and the derivative of $\Phi$ in $Q$ there is $v \mapsto 2\big((Q_* - A)\cdot v, (Q_* - C)\cdot v\big)$, with determinant $\pm 4\sin(\beta_U - \beta_D)$. This is bounded away from zero on $\mathcal{K}$.

The analytic implicit function theorem gives, around each point $(\beta, 0)$ with $\beta \in \mathcal{K}$, a neighbourhood and a radius $\rho_\beta$ such that $\Phi(Q, p) = 0$ has exactly one solution with $|Q - Q_*(\beta)| < \rho_\beta$, depending analytically on $p$. Finitely many of these neighbourhoods cover $\mathcal{K} \times \{0\}$. Where two overlap, the two solutions agree by uniqueness, provided they lie within the smaller radius, and this holds after shrinking the neighbourhoods, because $Q_*$ is uniformly continuous. So there are $c_1 > 0$ and one analytic branch $Q(\beta, c)$ on a neighbourhood of $\mathcal{K} \times [-c_1, c_1]$.

The quotient $f = X/U = (Q - A)/C$ is analytic there, since $|C| \ge \frac{1}{2}$. Taylor's theorem in $c$ at fixed $\beta$, applied to the real and imaginary parts, gives

$$f(\beta, c) = f(\beta, 0) + c\,\partial_c f(\beta, 0) + r_X, \qquad |r_X| \le c^2 \sup_{\mathcal{K}\times[-c_1,c_1]} |\partial_c^2 f|.$$

The supremum is finite by compactness. Here $f(\beta, 0) = 1$, and Lemma 3.2, applied to the family with fixed bearings, gives $\partial_c f(\beta, 0) = ie^{i\beta_D}$. The same argument applies to $Y/D = (C - Q)/(-A)$. For the triangle step, the normals $\omega U$ and $\bar\omega U$ are never parallel, so any compact set of $\beta_U$ works. The last sentence is the same argument with the corresponding $Q_*$. □

The coefficients are evaluated at the bearings current at each step, not at the initial ones. The analytic and compactness argument is not formalised in Lean.

## 4. The first disk near a hexagonal front

A rhombic lattice front whose notch hinge is exactly 60° is a piece of the hexagonal packing: every disk of the front touches its neighbours in the triangular lattice. This section shows that such fronts are unstable in the direction the conjecture cares about. Perturb a few hinges by amounts of order $t$ and let the disks shrink at a rate of order $t$, and the very first disk can be a pentagon.

### 4.1. **The notch-60 lattice.**

**Lemma 4.1** (Notch-60 law)**.** *A rhombic lattice front with $m$ down vectors and $n$ up vectors of bearing $u$, the down vectors at $u - 120°$ (notch hinge 60°), closes if and only if*

$$\tan u = \frac{\sqrt{3}\,m}{2n - m}.$$

*Both families point to the right, $30° < u < 90°$, exactly when $n/2 < m < 2n$.*

*Proof.* Expand $\sin(u-2\pi/3) = -\frac{1}{2}\sin u - \frac{\sqrt{3}}{2}\cos u$: the closing condition $m\sin(u-2\pi/3) + n\sin u = 0$ becomes $(2n-m)\sin u = \sqrt{3}\,m\cos u$. The bounds on $u$ are $\tan u > 1/\sqrt{3}$ and $2n-m>0$. □

For $(2,3)$ this gives $\tan u = \sqrt{3}/2$, $u \approx 40.89°$ and down bearing $\approx -79.11°$; for $(1,2)$ it gives a vertical down vector, which is why the construction below needs $m,n \geq 2$.

**Definition 4.2.** Let $m,n \geq 2$ with $n/2 < m < 2n$. Write the front word as $D_1\cdots D_m U_1\cdots U_n$ and name the disks around the notch $Z \xrightarrow{D_{m-1}} A \xrightarrow{D_m} B \xrightarrow{U_1} C \xrightarrow{U_2} E$. The front $H(\alpha,\sigma,\varepsilon,c)$ has all its disks, the seeds included, of diameter (2), consecutive disks touching, and the hinges of the notch-60 lattice except four:

$$h(D_{m-1},D_m) = 180° + \alpha, \qquad h(D_m,U_1) = 60° + \sigma, \qquad h(U_1,U_2) = 180° - \varepsilon,$$
$$h(U_n,D_1) = 300° - (\alpha+\sigma-\varepsilon).$$

The last hinge, at the ridge, is forced: the deviations $h-180°$ of the hinges of a closed front sum to zero. To fix scale and vertical translation, set the diameter of the first disk to 1 at height 0. The bearings have the form $\phi + \beta_j(\alpha,\sigma,\varepsilon)$, where the hinges determine the relative bearings $\beta_j$. Generate the edge lengths by (8). At zero parameters the vertical closing error has derivative $\sum_j \cos\beta_j = \Lambda_0 > 0$ with respect to $\phi$. The implicit function theorem therefore determines an analytic closing rotation for small parameters. Horizontal closure defines the positive circumference. The last tangency is included in this recurrence, since its endpoint returns to height 0 and hence to diameter 1. Both $\eta_U$ and $\eta_D$ are $O(|\alpha|+|\varepsilon|+|c|)$, so their maximum has this bound and their product has its square as a bound. The one notch angle is $60°+\sigma$, and the rate is $|c|$.

At $t=0$ the disks of $H$ sit on the triangular lattice $\{xD+yU : x,y\in\mathbb{Z}\}$, where $D$ and $U$ now denote the unit down and up vectors. Since $D\cdot U = -\frac{1}{2}$,

$$|xD+yU|^2 = x^2 - xy + y^2, \tag{9}$$

which takes the values $0,1,3,4,7,\ldots$ and never 2. Two disks of $H$ that are not neighbours are at distance at least $\sqrt{3}$ at $t=0$, except two pairs at distance 1: the disks on either side of the notch, $A$ and $C$, and the disks on either side of the ridge. Indeed, going right from one disk to the other along the front adds the vectors of a proper cyclic subword of $D^mU^n$, with sum $xD+yU$, $0\leq x\leq m$, $0\leq y\leq n$. By (9) this has length 1 only for $(x,y)$ equal to $(1,0)$ or $(0,1)$, which are neighbours, or $(1,1)$, which is the subword $DU$ at the notch or $UD$ at the ridge. The other images of the second disk add $k(mD+nU)$. For $k=-1$ the result is minus the sum of the complementary subword, covered by the same list. For $k\geq 1$ or $k\leq -2$ both coordinates are at least 2 or both at most $-2$, and since $x^2-xy+y^2 = \frac{1}{4}(x+y)^2 + \frac{3}{4}(x-y)^2$ the squared length is at least 4.

**Lemma 4.3** (Validity)**.** *Along $(\alpha,\sigma,\varepsilon,c) = t(a,s,e,-q)$ the clearances of these two pairs are, to first order, $\frac{\sqrt{3}}{2}\big(s+\frac{q}{2}\kappa_u\big)t$ at the notch and $\frac{\sqrt{3}}{2}\big(a+s-e-\frac{q}{2}\kappa_u\big)t$ at the ridge, with $\kappa_u = \cos(u-60°)$. So the lower bounds in* (1) *make $H(ta,ts,te,-tq)$ a valid front for small $t>0$.*

*Proof.* Let $T$ be the middle disk of either triple and $L$, $R$ its neighbours. Write $\ell_1 = |T-L|$, $\ell_2 = |T-R|$, and let $\phi$ be the angle of the triangle $LTR$ at $T$: the angle above the chain at the notch and below it at the ridge. Then $\phi$ is $60°$ plus a hinge deviation, which the definition prescribes exactly: $\phi = 60°+\sigma$ at the notch and $\phi = 60°+(\alpha+\sigma-\varepsilon)$ at the ridge. By the law of cosines,

$$|L-R|^2 = \ell_1^2 + \ell_2^2 - 2\ell_1\ell_2\cos\phi.$$

At $t=0$, $\ell_1=\ell_2=|L-R|=1$ and $\phi = 60°$, so $\frac{d}{dt}|L-R| = \frac{1}{2}(\dot\ell_1+\dot\ell_2) + \frac{\sqrt{3}}{2}\dot\phi$. Tangency gives $\ell_1 = \frac{1}{2}(d_T+d_L)$ and $\ell_2 = \frac{1}{2}(d_T+d_R)$ exactly. The clearance of the pair is $g = |L-R| - \frac{1}{2}(d_L+d_R)$, so

$$\dot g = \tfrac{\sqrt{3}}{2}\dot\phi + \tfrac{1}{2}\Big(\dot d_T - \tfrac{1}{2}(\dot d_L + \dot d_R)\Big).$$

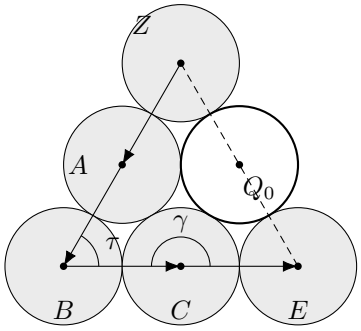



FIGURE 5. The notch of a hexagonal front, turned so that $U_1 = C - B$ is horizontal. The site $Q_0$ touches four front disks, $Z$, $A$, $C$ and $E$, and $Z$, $E$ are two diameters apart (dashed). The pentagon disk touches $A$ and $E$; at constant diameter it must clear $C$, which needs $\varepsilon > \sigma$, and fit between $Z$ and $E$, which needs $2\sigma + \alpha > \varepsilon$.

With $c = -qt$ and heights $y_k$ at $t = 0$, $\dot{d}_k = -q\,(y_k - y_0)$, and the reference height $y_0$ cancels. At the ridge $T$ is reached by $U_n$ and left by $D_1$, so $y_T - \frac{1}{2}(y_L + y_R) = \frac{1}{2}(\operatorname{Im} U - \operatorname{Im} D) = \frac{1}{2}\big(\sin u - \sin(u - 120°)\big) = \frac{\sqrt{3}}{2}\kappa_u$; at the notch the same expression is $-\frac{\sqrt{3}}{2}\kappa_u$. Hence

$$\dot{g}_{\text{notch}} = \tfrac{\sqrt{3}}{2}\big(s + \tfrac{q}{2}\kappa_u\big), \qquad \dot{g}_{\text{ridge}} = \tfrac{\sqrt{3}}{2}\big(a + s - e - \tfrac{q}{2}\kappa_u\big).$$

Every other pair of non-neighbours is at distance at least $\sqrt{3}$ at $t = 0$, by the remark before the lemma, so it stays clear for small $t$. Neighbours touch by construction. Under the lower bounds in (1) both derivatives are positive, so $g > 0$ at both pairs for small $t > 0$. □

Shrinking closes the ridge: the hexagonal front itself is not a valid front once $c < 0$, and the ridge has to open by at least $\frac{q}{2}\kappa_u t$. With the other hinges at their lattice values this forces $\alpha + \sigma - \varepsilon > 0$, so with $\sigma < \varepsilon$ the down vectors must be irregular too ($a > 0$). On fronts with $m + n \geq 5$ one can instead perturb another hinge inside a run, which changes nothing at $Q_0$.

Theorem A of the introduction says that $H(ta, ts, te, -tq)$ places a pentagon first when $(a, s, e, q)$ satisfies (1). The rest of this section proves it.

4.2. **The lowest site.**

**Lemma 4.4.** *At $\alpha = \sigma = \varepsilon = c = 0$, with unit diameters:*

(a) *every position touching two front disks is a lattice point;*
(b) *among lattice points strictly above the front, the lowest are $Q_0 = A + U$ and its periodic images, and every other one is higher by at least $\min(|\operatorname{Im} D|, \operatorname{Im} U)$;*
(c) *$Q_0$ touches exactly the four front disks $Z, A, C, E$; $|Z - E| = 2$ with $Q_0$ the midpoint, and $|Q_0 - B| = \sqrt{3}$.*

*Proof.* (a) A position touching disks at $P$ and $R$ has $|Q - P| = |Q - R| = 1$, so $|P - R| \leq 2$, and by (9) $|P - R|$ is 1, $\sqrt{3}$ or 2. The lattice is invariant under rotation by 60°. In the first case $Q = P + (R - P)e^{\pm i\pi/3}$. In the second, the six solutions of $x^2 - xy + y^2 = 3$ show that $R - P = w + we^{i\pi/3}$ for a unit lattice vector $w$, and the two positions are $P + w$ and $P + we^{i\pi/3}$. In the third, the solutions of $x^2 - xy + y^2 = 4$ are twice the unit vectors, and $Q$ is the midpoint. In each case $Q$ is a lattice point.

(b) Use coordinates $xD + yU$ and put $\delta = -\operatorname{Im} D > 0$, $\upsilon = \operatorname{Im} U > 0$. The height is $-x\delta + y\upsilon$, and closure of the period $mD + nU$ gives $m\delta = n\upsilon$. Take $B$ as the origin and $A = (-1, 0)$, so the front vertices are $(x, 0)$ for $-m \leq x \leq 0$ and $(0, y)$ for $0 \leq y \leq n$, repeated with period $(m, n)$. The map $(x, y) \mapsto xD + yU$ preserves orientation and $\operatorname{Re} D, \operatorname{Re} U > 0$, so the front is a graph over the horizontal axis, and a point lies above it exactly when it lies to the left of the staircase traversed in the direction of increasing $x$ and $y$. Write a lattice point as $(km - j,\, y)$ with $k \in \mathbb{Z}$ and $1 \leq j \leq m$. It is strictly above the front exactly when $y \geq kn + 1$. Relative to

$B$, and using $km\delta = knv$, its height is

$$-(km-j)\delta + yv = j\delta + (y-kn)v \;\geq\; j\delta + v.$$

This is smallest, $\delta + v$, exactly for $j = 1$, $y = kn+1$, which is $(-1,1) + k(m,n) = Q_0 + k(mD + nU)$. Any other point has $j \geq 2$ or $y \geq kn+2$, and so is higher by at least $\delta$ or $v$.

(c) The six unit neighbours of $Q_0 = (-1,1)$ are $Q_0 + (1,0) = (0,1) = C$, $Q_0 - (0,1) = A$, $Q_0 + (1,1) = (0,2) = E$, $Q_0 - (1,1) = (-2,0) = Z$, and $(-1,2)$, $(-2,1)$. The last two are not front vertices, because $m, n \geq 2$. Then $Z - E = -2(D+U)$ has length 2 with midpoint $Q_0$, and $Q_0 - B = U - D$ has length $\sqrt{3}$ by (9). □

So six pairs share the apex $Q_0$: $ZA$ (a triangle on $D_{m-1}$), $AC$ (the rhombus), $CE$ (a triangle on $U_2$), $AE$ and $ZC$ (pentagons) and $ZE$ (a transition removing three vectors). The first disk is decided among them at first order. The pair $ZE$ is degenerate and is treated by its separation, rather than by a transverse-apex expansion.

4.3. **First-order clearances.** Turn and translate the configuration so that $B = 0$ and $U_1$ points along 1; clearances do not change, and the height enters only multiplied by $c$, so its first-order change does not matter. Then $A = \omega$, $C = 1$, $E = 2$, $Z = 2\omega$, $Q_0 = 1 + \omega$ with $\omega = e^{i\pi/3}$, and the contact normals $n_k = Q_0 - z_k$ are

$$n_A = 1, \qquad n_C = \omega, \qquad n_E = \omega - 1, \qquad n_Z = 1 - \omega.$$

Along $(\alpha, \sigma, \varepsilon, c) = t(a, s, e, -q)$ write $\dot z$ for $dz/dt$ at $t = 0$. For a candidate family $Q(t) = Q_0 + tx + o(t)$, differentiating the clearance $|Q - z_k| - \frac{1}{2}(d_k + d_Q)$ as in (6) shows that, to first order, the disk is clear of $z_k$ to strict first order when $n_k \cdot x > \lambda_k$; a tangency requires $n_k \cdot x = \lambda_k$, with

$$\lambda_k = n_k \cdot \dot z_k - \tfrac{q}{2}\, h_k$$

and $h_k$ the height of $z_k$ above $Q_0$ (diameters normalised to 1 at the height of $Q_0$).

**Lemma 4.5.** *Put $\mu_C = \lambda_A + \lambda_E - \lambda_C$ and $\mu_Z = -(\lambda_Z + \lambda_E)$. To first order:*

(a) *the disk touching $A$ and $E$ clears $C$ by $\mu_C$ and $Z$ by $\mu_Z$;*
(b) *the disks touching $A, C$ and $C, E$ overlap $E$ and $A$ by $\mu_C$, and the disk touching $Z, C$ overlaps $A$ by $\mu_C + \mu_Z$;*
(c) *if $\mu_Z > 0$ no disk near $Q_0$ touches both $Z$ and $E$, and the disk touching $Z, A$ is higher than the one touching $A, E$ by $\frac{2}{\sqrt{3}}\mu_Z \cos u$;*
(d) *$\mu_C = \frac{\sqrt{3}}{2}(e - s) - \frac{q}{8}\big(3\sin u + 5\sqrt{3}\cos u\big)$ and $\mu_Z = \frac{\sqrt{3}}{2}(2s + a - e) + \frac{q}{2}\big(3\sin u + \sqrt{3}\cos u\big)$, per unit $t$.*

*Proof.* Parts (a) to (c) use only $n_E = n_C - n_A$ and $n_Z = -n_E$. At the vertex $n_A \cdot x = \lambda_A$, $n_E \cdot x = \lambda_E$ one gets $n_C \cdot x - \lambda_C = \lambda_A + \lambda_E - \lambda_C$ and $n_Z \cdot x - \lambda_Z = -\lambda_E - \lambda_Z$. At $n_A \cdot x = \lambda_A$, $n_C \cdot x = \lambda_C$ the clearance to $E$ is $\lambda_C - \lambda_A - \lambda_E = -\mu_C$, and symmetrically for $C, E$ against $A$. At $n_Z \cdot x = \lambda_Z$, $n_C \cdot x = \lambda_C$ the clearance to $A$ is $\lambda_Z + \lambda_C - \lambda_A = -\mu_C - \mu_Z$.

The pair $ZE$ has opposite normals, so it has no transverse apex and needs a separate argument. Suppose a disk at $Q$ touches both, with $Q \to Q_0$ as $t \to 0$. By the triangle inequality

$$|Z - E| \leq |Q - Z| + |Q - E| = \tfrac{1}{2}(d_Z + d_E) + d_Q,$$

so $g(t) = |Z - E| - \frac{1}{2}(d_Z + d_E) - d_Q \leq 0$. Normalise diameters to 1 at the height of $Q_0$ at $t = 0$; then $d_Q = 1 + c\,(\operatorname{Im} Q - \operatorname{Im} Q_0) = 1 + o(t)$. Since $Z - E = -2n_Z$ with $|n_Z| = 1$ at $t = 0$,

$$g(t) = t\big(-n_Z \cdot \dot Z - n_E \cdot \dot E + \tfrac{q}{2}(h_Z + h_E)\big) + o(t) = t\,\mu_Z + o(t),$$

using $n_E = -n_Z$ and $\dot d_k = -q h_k$. So $\mu_Z > 0$ gives $g(t) > 0$ for small $t > 0$, a contradiction. No apex expansion is used, so this covers candidates whose distance from $Q_0$ is much larger than $t$. The vertices $ZA$ and $AE$ both lie on the line $n_A \cdot x = \lambda_A$, a distance $\frac{2}{\sqrt{3}}\mu_Z$ apart along $in_A$, whose height in the cylinder frame is $\cos u > 0$.

(d) The hinges give $\dot A = is\,\omega$, $\dot C = 0$, $\dot E = ie$, $\dot Z = i(2s + a)\omega$ at $q = 0$, so $\lambda_A = -\frac{\sqrt{3}}{2}s$, $\lambda_C = 0$, $\lambda_E = \frac{\sqrt{3}}{2}e$ and $\lambda_Z = -\frac{\sqrt{3}}{2}(2s + a)$. The rate stretches each side to the mean of its two

diameters, adding $-q\ell_{BC}$ to $\dot{C}$, $-q(\ell_{BC}+\ell_{CE})$ to $\dot{E}$, $-q\ell_{AB}\omega$ to $\dot{A}$ and $-q(\ell_{AB}+\ell_{ZA})\omega$ to $\dot{Z}$, where $\ell_{XY}=\frac{1}{2}(h_X+h_Y)$. Collecting terms,

$$\mu_C^q=-\tfrac{q}{4}\,(3h_A-h_B-5h_C+h_E),\qquad \mu_Z^q=-\tfrac{q}{4}\,(2h_A+2h_B+2h_C-h_Z-h_E),$$

and the heights are $h_z=-\operatorname{Im}\bigl(e^{iu}(Q_0-z)\bigr)$: $h_A=-\sin u$, $h_B=-\sqrt{3}\sin(u+30^\circ)$, $h_C=-\sin(u+60^\circ)$, $h_E=-\sin(u+120^\circ)$, $h_Z=-\sin(u-60^\circ)$. Since $\sin(u-60^\circ)+\sin(u+120^\circ)=0$ this gives (d). □

*Proof of Theorem A.* By Lemma 4.3 the front is valid for small $t>0$. By (d) the upper bound in (1) is $\mu_C>0$. Since $\frac{1}{2}(3\sin u+\sqrt{3}\cos u)=\sqrt{3}\kappa_u$, (d) and Lemma 4.3 give $\mu_Z$ as the sum of the notch and ridge clearance rates plus $\sqrt{3}q\kappa_u$, so $\mu_Z>0$ on every valid front.

*The pentagon disk is a valid position.* The normals $n_A$ and $n_E$ are not parallel, so Lemma 3.1 gives a $C^1$ family $Q_{AE}(t)\to Q_0$ touching $A$ and $E$. By Lemma 4.5(a) its clearances from $C$ and $Z$ are $t\mu_C+o(t)$ and $t\mu_Z+o(t)$, both positive for small $t$. Every other front disk, periodic images included, is at distance at least $\sqrt{3}$ from $Q_0$ at $t=0$ (Lemma 4.4(c) and (9)), so it stays clear. $Q_0$ is strictly above the front, and the front moves continuously with $t$, so $Q_{AE}(t)$ is above it.

*Nothing else is as low.* Suppose that for some sequence $t_k\to 0$ there is a valid position $Q_k\neq Q_{AE}(t_k)$ with $\operatorname{Im}Q_k\le\operatorname{Im}Q_{AE}(t_k)$, touching front disks $P_k$ and $R_k$. We derive a contradiction in three steps.

*Step 1: compactness.* Normalise the diameter to 1 at height 0, so the front disks have diameters in $[\frac{1}{2},2]$ and heights in a fixed bounded interval for small $t$. If $Q$ touches $P$, then $|\operatorname{Im}Q-\operatorname{Im}P|\le|Q-P|=d_P+\frac{c}{2}(\operatorname{Im}Q-\operatorname{Im}P)$, so $|\operatorname{Im}Q-\operatorname{Im}P|\le 2d_P\le 4$ once $|c|\le\frac{1}{2}$. Hence $Q_k$ has bounded height, and modulo the period it lies in a bounded set. Its parents lie within distance 4 of it, and the circumference is bounded below, so up to periodic translation $P_k$ and $R_k$ range over a finite set of front disks and images. Passing to a subsequence, the pair $(P_k,R_k)$ is fixed and $Q_k\to Q^*$.

*Step 2: the limit is $Q_0$.* Touching and non-overlap are closed conditions, and all diameters tend to 1. So $Q^*$ is at distance exactly 1 from the limits of $P_k$ and $R_k$, and at distance at least 1 from every front disk at $t=0$. By Lemma 4.4(a), $Q^*$ is a lattice point. It lies on or above the limiting front. The only lattice points on the front polyline are its vertices, which are front disks at distance 0, so $Q^*$ is strictly above. Finally $\operatorname{Im}Q^*\le\lim\operatorname{Im}Q_{AE}(t_k)=\operatorname{Im}Q_0$. By Lemma 4.4(b), $Q^*$ is $Q_0$ up to periodic translation, and by Lemma 4.4(c) the limiting parents are two of $Z,A,C,E$.

*Step 3: the six pairs.* For large $k$, $Q_k$ is a position near $Q_0$ touching one of the six pairs. Five of them have non-parallel normals at $Q_0$. For these, the implicit function theorem in the proof of Lemma 3.1, with $t$ in place of $c$, gives a neighbourhood of $Q_0$ in which the touching position is unique for small $t$ and lies on the $C^1$ apex family, so $Q_k$ lies on that family.

- $AE$: then $Q_k=Q_{AE}(t_k)$, which was excluded.
- $AC$, $CE$ and $ZC$: by Lemma 4.5(b) these families overlap $E$, $A$ and $A$ by $t\mu_C+o(t)$, $t\mu_C+o(t)$ and $t(\mu_C+\mu_Z)+o(t)$, so they are not valid for large $k$.
- $ZA$: by Lemma 4.5(c) it is higher than $Q_{AE}$ by $\frac{2}{\sqrt{3}}\mu_Z\cos u\,t+o(t)>0$.

The sixth pair, $ZE$, admits no touching position near $Q_0$, by the separation argument in the proof of Lemma 4.5. Every case gives a contradiction. So for small $t>0$ the pentagon disk touching $A$ and $E$ is the unique lowest valid position, and the first transition is a pentagon. □

Lemma 4.5, the notch part of Lemma 4.3 and the step from validity to $\mu_Z>0$ are checked in Lean (`Static.lean`): the four tangencies and three hinge rates of the front determine the first-order clearances of all six candidates and of the notch pair. The file `StaticExists.lean` constructs the local five-disk and apex families for the prescribed rates. Periodic closure and global exclusion are supplied by the analytic arguments above.

On valid rays above the window ($\mu_C<0$), the tested fronts place a rhombus $AC$. The independent simulator agrees with both predictions on every valid front tested: 182 fronts over all

26 shapes with $2 \le m \le 6$, on rays inside and above the window (`verify/static_general.py`). A separate brute-force enumeration confirms Lemma 4.4(b) for all 47 shapes with $2 \le m \le 8$ (`verify/static_lattice_order.py`).

The theorem uses the simplest perturbation. Lemma 4.5 involves only the hinges at $A$, $B$ and $C$, so on fronts with $n \ge 3$ one may also turn the hinge between $U_2$ and $U_3$, and then $\alpha = 0$ is allowed: the down vectors stay exactly regular and only the up vectors carry irregularity. The channel condition $\mu_Z > 0$ is then no longer automatic. When it fails the channel between $Z$ and $E$ is closed, no disk fits near $Q_0$, and the first disk is placed elsewhere, a triangle in all our runs (`verify/static_channel.py`).

## 5. Front shape is frozen during rhombic phases

A *rhombic phase* is a run of $L$ consecutive rhombus steps. In each step an up vector passes the down vector on its left (Figure 4). Lemma 3.2 says that the factor an up vector receives depends only on the down vector it passes. If every up vector passed every down vector equally often and at the same times, all up vectors would receive the same product of factors, and their ratios would not move. The bookkeeping below shows that this is true up to boundedly many factors, whatever the phase length.

5.1. **Pass bookkeeping.** Number the $n$ up vectors in their cyclic order, which never changes, and let $g_i$ be the number of down vectors between up $i$ and up $i+1$. Let $P_i(t)$ be the number of passes made by up $i$ in the first $t$ steps. A pass by up $k$ moves one down vector from gap $k-1$ to gap $k$, so

$$g_i(t) = g_i(0) + P_i(t) - P_{i+1}(t). \tag{10}$$

**Lemma 5.1** (Pass balance, lag and sweep)**.** *In a rhombic phase with $m$ down and $n$ up vectors:*

(a) $|P_j(t) - P_k(t)| \le m$ *for all ups $j, k$ and all $t$;*
(b) *if $t_{u,k}$ is the step of the $k$-th pass of up $u$, then $|nk - t_{u,k}| \le nm$;*
(c) *labelling the down vectors $0, \dots, m-1$ in their cyclic order, up $u$ passes down vector $\alpha_u - k \pmod m$ at its $k$-th pass, for a fixed $\alpha_u$.*

*Proof.* (a) Summing (10) over the gaps from up $j$ to up $k$ gives $P_j - P_k$ as the change in the number of down vectors between them, which lies in $[-m, m]$. (b) The $n$ counts sum to $t$, so by (a) each lies within $m$ of $t/n$; apply this at $t = t_{u,k}$, where $P_u = k$. (c) Down vectors never pass each other, so they keep their cyclic labels. Let $r_u$ be the label of the first down vector to the right of up $u$. A pass by $u$ moves the down vector labelled $r_u - 1$ to its right, so $r_u$ drops by one; a pass by another up changes neither the order of the down vectors nor which one is first to the right of $u$. Consistency of the labels with the gaps, $r_{i+1} = r_i + g_i$, is preserved by every pass. □

5.2. **The frozen-shape proposition.**

**Proposition 5.2.** *Consider a rhombic phase with $m$ down and $n$ up vectors in which every step obeys Lemma 3.2 with normalised multiplicative remainder at most $Rc^2$, and in which the phase $e^{i\beta_d(t)}$ of each down vector changes by at most $\lambda$ per step. If $|c| + Rc^2 \le \frac{1}{2}$, then for any two up vectors $u, v$ making $K_u$ and $K_v$ passes,*

$$\left|\Delta \log \frac{U_u}{U_v}\right| \le |c|\big(3nm\,\lambda\,K_u + 3m\big) + (K_u + K_v)\big((|c| + Rc^2)^2 + Rc^2\big),$$

*where $\Delta \log$ is the accumulated change of the logarithm (the continuous branch along the phase). The same holds for two down vectors with $m$ and $n$ exchanged.*

With $\lambda = O(|c|)$ (each vector changes by $O(c)$ in the steps that involve it) and $K_u \le L/n + m$, the bound is $O_{m,n}(|c| + c^2 L)$. For phases of length $O(1/|c|)$, every up/up and down/down ratio of the front, angle and length, changes by $O(|c|)$. The bound needs no smallness of the irregularity, only the compactness of Lemma 3.5.

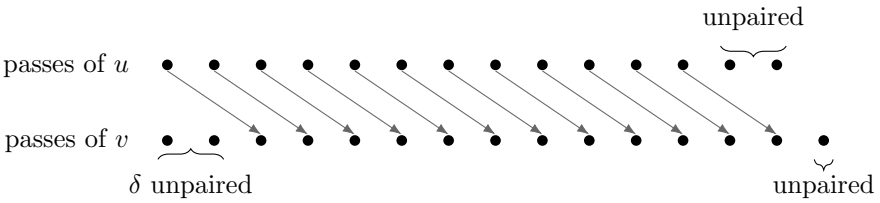
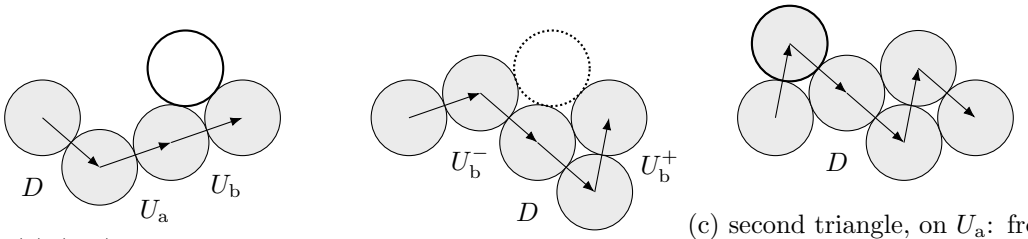



FIGURE 6. The matching in the proof of Proposition 5.2 with $\delta = 2$. Each paired pass meets the same down vector at times at most $3nm$ apart; at most $3m$ passes are left unpaired.




(a) $(1,2)$ at threshold: first triangle on $U_{\mathrm{b}}$

(b) $(2,2)$ phase: the triple $U_{\mathrm{b}}^-, D, U_{\mathrm{b}}^+$ (margin $\sqrt{7}|c|$)

(c) second triangle, on $U_{\mathrm{a}}$: front $(3,2)$ with steep ups $U_{\mathrm{a}}^+$, $U_{\mathrm{b}}^+$

FIGURE 7. The first Fibonacci transition at the $(1,2)$ threshold tiling (exact coordinates; open disks are new triangle disks). In (b) the dotted disk marks where a pentagon would sit: at the limit it touches three disks, and with shrinking disks it misses by $\sqrt{7}|c|$ to first order.

*Proof.* By Lemma 3.2, up $u$ is multiplied at its $k$-th pass by $1 + ica_k + r_k$ with $a_k = e^{i\beta_{d(u,k)}(t_{u,k})}$ and $|r_k| \le Rc^2$; likewise $1 + icb_k + s_k$ for $v$. For $|z| \le \frac{1}{2}$, $|\log(1+z) - z| \le |z|^2$, so the change of $\log(U_u/U_v)$ is $ic\big(\sum_{k<K_u} a_k - \sum_{k<K_v} b_k\big)$ up to the second term of the bound.

By Lemma 5.1(c), the $k$-th pass of $u$ and the $(k+\delta)$-th pass of $v$ meet the same down vector, where $\delta = (\alpha_v - \alpha_u) \bmod m < m$. Pair them (Figure 6). By (b), their times differ by at most $nm + nm + n\delta \le 3nm$, so the paired phases differ by at most $3nm\,\lambda$. Write

$$\sum_{k<K_u} a_k - \sum_{k<K_v} b_k = \sum_{k<K} (a_k - b_{k+\delta}) + \sum_{K\le k<K_u} a_k - \sum_{j<\delta} b_j - \sum_{K+\delta\le j<K_v} b_j,$$

with $K = \min(K_u, K_v - \delta)$. The first sum is at most $3nm\,\lambda K_u$. The other three contain $(K_u - K) + \delta + (K_v - K - \delta) \le |K_u - K_v| + 2\delta \le 3m$ terms of modulus one, by (a). If $K_v < \delta$, every term is unpaired and there are $K_u + K_v < 3m$ of them. □

This matching proof is simpler than the cycle-by-cycle regrouping one might first try: it needs one shift, by the offset $\delta$ of the two sweeps, and no complete cycles. Atela and Golé showed that at constant diameter the tangential eigenvalues of the stacking map at a rhombic tiling are roots of unity [3], so a priori a slow forcing could build up resonantly over a phase. Proposition 5.2 shows that the first-order forcing cancels instead.

## 6. The first transition

Figure 7 shows the whole first transition at its limit, $\varepsilon = c = 0$: a triangle on one up vector, a $(2,2)$ phase in which one triple of disks comes close to a pentagon without reaching it, and a triangle on the other up vector.

6.1. **The $(1,2)$ phase.** A front with one down vector has a single notch, $D$ followed by an up vector, so in a rhombic phase the two up vectors pass $D$ alternately. Each pass turns $D$ by $-c\cos\beta_U > 0$ and the passing up vector by $c\cos\beta_D < 0$ (Corollary 3.4), so the notch hinge $\pi - (\beta_U - \beta_D)$ increases steadily, while by Proposition 5.2 the ratio $U_{\mathrm{a}}/U_{\mathrm{b}}$ stays $e^{i\varepsilon}$ up to $O(|c|)$.

Along the way the configuration follows the $(1,2)$ lattice fronts from $F(\beta,\varepsilon)$ to the threshold of Lemma 2.2, up to $O(|\varepsilon|+|c|)$. While the notch hinge stays in a fixed compact subset of $(60^\circ,120^\circ)$, the 1, 2-notch has $\gamma+\tau>4\pi/3$ by a margin bounded below, so by [5, Prop. 4.3] and Lemma 3.1 the only valid offspring is the rhombus. Near $120^\circ$ the configuration is close to the threshold tiling, where Lemma 6.1(b) applies instead: the rhombus remains the valid candidate at the notch until the up triangle becomes valid.

The phase ends at the first step at which a triangle candidate is valid. At the threshold tiling this is a triple site: the triangle on $D$, the triangle on the up vector that follows it, and the rhombus over the notch disk are one and the same disk, touching three disks. So the switch does not happen at a hinge of exactly $120^\circ$ once $c\neq 0$ (it happens at $120^\circ+O(|c|)$), and which triangle is placed is decided at first order, not by an order-one margin. Lemma 6.1 settles it: whenever the triangle on $D$ is valid, the triangle on the up vector is valid and no higher, with equality only when the two are the same disk, which is then inserted on the up vector. So the first triangle rests on an up vector. Call it $U_{\mathrm{b}}$ and the other $U_{\mathrm{a}}$. (When $|c|\ll|\varepsilon|$, $U_{\mathrm{b}}$ is the less steep one; Theorem B does not need that.)

6.2. **Two candidates sharing a contact.** Every near-tie in this section has the same shape. Put $u_*=\beta_{\min}=\arctan(\sqrt{3}/5)$ and let

$$U=e^{iu_*},\qquad S=e^{i(u_*-\pi/3)},\qquad V=e^{i(u_*+\pi/3)}$$

be the unit up, down and steep up vectors of the $(1,2)$ threshold tiling (Lemma 2.2).

**Lemma 6.1** (Shared contact)**.** *Let $A$, $B$, $E$ be disks of a front and $Q_*$ a point at which, at $\varepsilon=c=0$, a unit disk touches all three, with contact normals*

$$Q_*-A=U,\qquad Q_*-B=V,\qquad Q_*-E=-S.$$

*Suppose $A$ is a highest disk of the front. For configurations in a small enough neighbourhood of this one (positions, diameters, and $|c|$ small), let $T$, $R$ and $P$ be the placements touching $A$ and $B$, $B$ and $E$, and $A$ and $E$, on the branches through $Q_*$. Then:*

(a) *if $T$ clears $E$, then $R$ clears $A$ and $\operatorname{Im}R\le\operatorname{Im}T$, with equality only if $T=R$;*
(b) *$P$ clears $B$ if and only if $R$ does not clear $A$ strictly; so $P$ and $R$ are not both valid except when $P=R=T$, and one of them always clears the third disk;*
(c) *every other disk of the stack, older disks included, clears all three.*

*Proof.* For a disk $P$ write $f_P(X)=|X-P|-\frac12\big(d_P+d(X)\big)$, with $d(X)=b_0+c\operatorname{Im}X$; $X$ clears $P$ when $f_P(X)\ge 0$. Near $Q_*$ the curve $f_B=0$ is a graph $y=h(x)$, since its $y$-derivative is close to $\operatorname{Im}V>0$, and $h'=-\operatorname{Re}V/(\operatorname{Im}V-c/2)+o(1)<0$ because both coordinates of $V$ are positive. Along it put $g_A(x)=f_A(x,h(x))$ and $g_E(x)=f_E(x,h(x))$. At the limit

$$g_A'=\frac{\sin(\pi/3)}{\sin(u_*+\pi/3)}>0,\qquad g_E'=\frac{\sin(-\pi/3)}{\sin(u_*+\pi/3)}<0,$$

and both signs persist nearby. $T$ is the zero $x_T$ of $g_A$ and $R$ the zero $x_R$ of $g_E$, unique by the implicit function theorem since the contact normals are not parallel. If $T$ clears $E$ then $g_E(x_T)\ge 0=g_E(x_R)$, so $x_T\le x_R$ and $\operatorname{Im}T=h(x_T)\ge h(x_R)=\operatorname{Im}R$; and $g_A(x_R)\ge g_A(x_T)=0$, so $R$ clears $A$. Equality forces $x_T=x_R$. This is (a).

For (b), run the same argument on the curve $f_E=0$ through $Q_*$, a graph over the direction $t=e^{i(u_*+30^\circ)}$ perpendicular to $Q_*-E=-S$. Along it $f_A$ and $f_B$ both increase, with derivatives $U\cdot t=V\cdot t=\cos 30^\circ>0$ at the limit, persisting nearby. $P$ is the zero of $f_A$ and $R$ the zero of $f_B$ on this curve. So $f_B(P)\ge 0$ if and only if $P$ is not before $R$ along $t$, if and only if $f_A(R)\le f_A(P)=0$. Equality in either is the triple point.

For (c): before the first offending event every step is a triangle (which removes nothing) or a rhombus or pentagon (which removes disks lower than the new disk), so by induction no centre in the stack is higher than the highest front disk, which is $A$, at height $\operatorname{Im}Q_*-\sin u_*+o(1)$, below $Q_*$. The directions from $Q_*$ to $A$, $B$, $E$ are $u_*+180^\circ$, $u_*+240^\circ$, $u_*+300^\circ$, about $199^\circ$, $259^\circ$, $319^\circ$, and every direction in the lower half-plane is within less than $60^\circ$ of one of them. A unit disk centred at $Q_*+rn$, $0<r\le 1$, $n$ in the lower half-plane, would therefore lie at distance

$\sqrt{1+r^2-2r\cos\theta}<1$ from that parent, $\theta<60°$, overlapping it. So at the limit no other disk meets $Q_*$; any disk overlapping a nearby candidate $X$ has its centre within $d(X)/(1-|c|/2)$ of $X$, so by compactness the exclusion persists in a neighbourhood. □

In words: both candidates touch $B$, and the curve of such positions descends to the right. Clearing the right-hand disk $E$ forces $T$ onto the higher part of that curve. The lemma is used twice. In a down-up notch $A \xrightarrow{S} B \xrightarrow{U} E$ ($E = B + U$), $T$ is the triangle on the down vector, $R$ the triangle on the up vector and $P$ the rhombus over $B$; by (b) the rhombus is valid exactly until the up triangle is, and by (a) the down triangle never undercuts it. In every use $A$ is the highest front disk: it is the only peak of the front in the $(1,2)$ phase and in the words $U_\mathrm{a}\,U_\mathrm{b}^-\,D\,U_\mathrm{b}^+$ and $U_\mathrm{a}\,D\,U_\mathrm{b}^-\,U_\mathrm{b}^+$, and in the other two words it exceeds the other peak by $\mathrm{Im}(S+V)=\sin u_*>0$ up to $O(\delta+|c|)$. In the words $U_\mathrm{a}\,U_\mathrm{b}^-\,D\,U_\mathrm{b}^+$ and $U_\mathrm{a}\,D\,U_\mathrm{b}^-\,U_\mathrm{b}^+$ below, $A, B$ are the ends of the first down vector and $E$ the end of the up vector $U_\mathrm{b}^+$, so $T$ is a triangle on a down vector and $R$ the rhombus. The comparison is exact; it does not need a margin.

6.3. **The** $(2,2)$ **phase.** After the first triangle the front is $U_\mathrm{a}, D, U_\mathrm{b}^+, U_\mathrm{b}^-$, where $U_\mathrm{b}^+$ and $U_\mathrm{b}^-$ are the new up and down vectors. At $\varepsilon=c=0$ these are the four vectors of the $(1,2)$ threshold tiling: $U_\mathrm{a}$ at $\beta_{\min}$, $D$ and $U_\mathrm{b}^-$ both at $\beta_{\min}-60°$, and $U_\mathrm{b}^+$ at $\beta_{\min}+60°$. The phase runs through four words, read cyclically from $U_\mathrm{a}$:

$$U_\mathrm{a}\,D\,U_\mathrm{b}^+\,U_\mathrm{b}^- \;\to\; U_\mathrm{a}\,U_\mathrm{b}^-\,D\,U_\mathrm{b}^+ \;\to\; U_\mathrm{a}\,U_\mathrm{b}^-\,U_\mathrm{b}^+\,D \;\to\; U_\mathrm{a}\,D\,U_\mathrm{b}^-\,U_\mathrm{b}^+ \;\to\; \cdots,$$

in which $U_\mathrm{a}$ and $U_\mathrm{b}^+$ pass strictly alternately.

**Lemma 6.2.** *In each of the four words at the limit tiling, the lowest valid position is unique up to the degenerate coincidences listed below, and every valid position at another location is at least* $\sqrt{3/28}\,b_0\approx 0.327\,b_0$ *higher. Consequently the passes alternate for* $|\varepsilon|,|c|$ *small, and the gap between* $U_\mathrm{a}$ *and* $U_\mathrm{b}^+$ *is* $1+P_{U_\mathrm{a}}-P_{U_\mathrm{b}^+}\in\{1,2\}$*; it is never empty.*

*Proof.* Each of the four words is a word in the vectors of the $(1,2)$ threshold tiling, so in the rational model with 7 in place of 19 every candidate position is a rational point: pairs at distance 1, $\sqrt{3}$ and 2 have their apexes at lattice points and farther pairs have none. The candidates were enumerated and compared in exact rational arithmetic (`verify/exact_limits.py`; an independent interval-arithmetic check is `verify/certify_limits.py`). The coincidences are exact identities of rational points, of two kinds: in the first and third words a triangle on a down vector, the triangle on $U_\mathrm{a}$ and a rhombus; in the second and fourth a triangle on a down vector, a rhombus and a pentagon. Both are treated below. The value $\sqrt{3/28}$ is exact: in the word $U_\mathrm{a}\,D\,U_\mathrm{b}^+\,U_\mathrm{b}^-$ the lowest position is the apex $w_3+U_\mathrm{a}$ of the triangle on $(U_\mathrm{b}^-,U_\mathrm{a})$ and the next is the rhombus apex $w_1+U_\mathrm{b}^+$, half a unit of $Y$ higher in the rational model. The gap formula is (10) applied to alternating passes. □

6.4. **No pentagon in the first transition.** A pentagon needs a three-vector notch whose two hinges sum to at most 240°. Three such triples are degenerate at zeroth order in the $(2,2)$ phase.

The triple $D, U_\mathrm{a}, U_\mathrm{b}^+$ (one down, two ups) needs $U_\mathrm{a}$ immediately followed by $U_\mathrm{b}^+$, an empty gap, which Lemma 6.2 excludes.

The other two have two downs and one up, namely $U_\mathrm{b}^-, D, U_\mathrm{b}^+$ in the word $U_\mathrm{a}\,U_\mathrm{b}^-\,D\,U_\mathrm{b}^+$ and $D, U_\mathrm{b}^-, U_\mathrm{b}^+$ in the word $U_\mathrm{a}\,D\,U_\mathrm{b}^-\,U_\mathrm{b}^+$. By (3) the hinge sum minus 240° telescopes to

$$m_j=\tfrac{2\pi}{3}-\big(\arg U_\mathrm{b}^+-\arg d_j\big),$$

where $d_j$ is the first down vector of the triple ($U_\mathrm{b}^-$, respectively $D$); the middle vector drops out. Write $u$ for the bearing of $U_\mathrm{b}$ when it is split, $a$ for that of $U_\mathrm{a}$ and $d$ for that of $D$, all within $\delta$ of their threshold values, and use the two first-order rules: a triangle on $U_\mathrm{b}$ creates $\arg U_\mathrm{b}^+=u+\frac{\pi}{3}+\frac{c}{2}\cos(u-\frac{\pi}{3})$ and $\arg U_\mathrm{b}^-=u-\frac{\pi}{3}-\frac{c}{2}\cos(u+\frac{\pi}{3})$ (Lemma 3.3), and a pass of an up bearing $a$ over a down bearing $d$ sends $a\mapsto a+c\cos d$, $d\mapsto d-c\cos a$ (Corollary 3.4), each up to $O(c^2)$.

For $d_j = U_{\mathrm{b}}^-$: the triple first forms after $U_{\mathrm{a}}$ passes $U_{\mathrm{b}}^-$, so

$$m_{U_{\mathrm{b}}^-} = -c\big(\tfrac{1}{2}\cos u + \cos a\big) + O(\delta|c| + c^2) = \tfrac{15}{4\sqrt{7}}|c| + O(\delta|c| + c^2) \approx 1.417\,|c|, \tag{11}$$

with $\cos\beta_{\min} = 5/(2\sqrt{7})$. For $d_j = D$: put $\eta_0 = \frac{\pi}{3} - (u - d)$ when the first triangle is placed. That triangle clears its left neighbour, and the clearance is $\frac{\sqrt{3}}{2}\eta_0 - \frac{3}{4}c\sin(u_* + \frac{\pi}{3}) + O(\eta_0^2 + \delta|c| + c^2)$ with root exactly $\eta_0 = 0$ at $c = 0$, so $\eta_0 \geq \frac{9}{4\sqrt{7}}c + O(\delta|c| + c^2)$. Before the triple forms, $U_{\mathrm{b}}^+$ and then $U_{\mathrm{a}}$ have passed $D$, so

$$m_D = \eta_0 - c\big(\tfrac{1}{2}\cos s_* + \cos s_* + \cos v_* + \cos u_*\big) + O(\delta|c| + c^2) \geq \tfrac{15}{4\sqrt{7}}|c| + O(\delta|c| + c^2), \tag{12}$$

since the cosine sum is $6/\sqrt{7}$ (here $s_* = u_* - \frac{\pi}{3}$, $v_* = u_* + \frac{\pi}{3}$).

Later passes only increase either margin: every rhombus that changes $m_j$ decreases $\arg U_{\mathrm{b}}^+$ or increases $\arg d_j$, by $|c|$ times a cosine bounded away from zero, up to $O(c^2)$. So no unsigned error accumulates over the phase, however long it is, and both margins stay at least $\frac{15}{4\sqrt{7}}|c|$ up to $O(\delta|c| + c^2)$. A pentagon needs more than a negative margin: reflecting centres by $z \mapsto -\bar{z}$ and reversing the front order (so front vectors become $v \mapsto \bar{v}$) turns either triple into a one-down-two-up notch whose down vector has $\cos = 1/(2\sqrt{7})$, $\sin = -3\sqrt{3}/(2\sqrt{7})$, and Lemma 7.2 below then requires the margin to be below $-\kappa_2|c|$ with $\kappa_2 = 13/(4\sqrt{7}) \approx 1.228$. The two requirements are separated by

$$\tfrac{15}{4\sqrt{7}} + \tfrac{13}{4\sqrt{7}} = \sqrt{7} \approx 2.646$$

times $|c|$, so no pentagon forms.

6.5. **No triangle on a down vector.** In the words $U_{\mathrm{a}}\,U_{\mathrm{b}}^-\,D\,U_{\mathrm{b}}^+$ and $U_{\mathrm{a}}\,D\,U_{\mathrm{b}}^-\,U_{\mathrm{b}}^+$ the exact tables put a triangle on the first down vector, the rhombus on the next two vectors and the pentagon of the triple above at one point (`verify/exact_limits.py`). The pentagon is excluded, and by Lemma 6.1 the rhombus is valid and no higher whenever the triangle is valid. In fact the gap is strict along the trajectory: expanding both placements to first order gives, with $b$ the diameter of the first disk of the triple,

$$\begin{aligned}\frac{\operatorname{Im} T - \operatorname{Im} R}{b} &= m_j\Big(\frac{1}{2\sqrt{7}} + O(\delta)\Big) - c\Big(\frac{13}{56} + O(\delta)\Big) + O(c^2)\\ &\geq |c|\Big(\frac{15}{56} + \frac{13}{56}\Big) - O(\delta|c| + c^2) = \frac{|c|}{2} - O(\delta|c| + c^2),\end{aligned}$$

using (11) and (12). In simulations the smallest observed value of $(\operatorname{Im} T - \operatorname{Im} R)/(b|c|)$ is 0.4997 (`verify/down_triangle_exclusion.py`).

*Proof of Theorem B.* In the $(1, 2)$ phase the only offspring is the rhombus until a triangle first becomes valid, at a triple site where Lemma 6.1 places it on an up vector. In the $(2, 2)$ phase the passes alternate with the margin of Lemma 6.2, and the coincidences at the lowest location are of two kinds. In the words $U_{\mathrm{a}}\,D\,U_{\mathrm{b}}^+\,U_{\mathrm{b}}^-$ and $U_{\mathrm{a}}\,U_{\mathrm{b}}^-\,U_{\mathrm{b}}^+\,D$ a triangle on the down vector before $U_{\mathrm{a}}$ meets the triangle on $U_{\mathrm{a}}$ and a rhombus, at the down-up notch before $U_{\mathrm{a}}$. By Lemma 6.1 for that notch no triangle on the down vector is placed; the triangle on $U_{\mathrm{a}}$, when valid, is the intended second triangle, and at an exact tie the same disk is inserted on $U_{\mathrm{a}}$. In the other two words a triangle on a down vector meets the rhombus and a pentagon; the pentagon is excluded and the rhombus is placed, as shown above. The remaining comparisons hold at the limit tilings with margins of order one.

It remains to see that the configuration stays in the neighbourhood $\mathcal{N}$ of the limit tilings on which all these statements and the rates of Corollary 3.4 hold, with constants uniform on $\mathcal{N}$. At the start of the $(2, 2)$ phase it is within $C_0(|\varepsilon| + |c|)$ of the limit, by Proposition 5.2 and Lemma 3.3. The phase ends when the notch $(U_{\mathrm{b}}^-, U_{\mathrm{a}})$ admits its triangle; that notch's hinge starts within $C_0(|\varepsilon| + |c|)$ of its switching value, and while the configuration is in $\mathcal{N}$ every pass of $U_{\mathrm{a}}$ over $U_{\mathrm{b}}^-$ moves it towards that value by at least $\kappa|c|$, $\kappa > 0$, since both vectors turn with the signed rates of Corollary 3.4. Let $n_1$ be the first step at which the configuration leaves the ball of radius $r$ about the limit, $r$ fixed with that ball inside $\mathcal{N}$. Before $n_1$ the phase has at

most $C_1(|\varepsilon|/|c|+1)$ passes, each of a bounded number of steps, so the configuration moves by at most $C_2(|\varepsilon|+|c|)$ before the phase ends; for $|\varepsilon|$ and $|c|$ small this is less than $r - C_0(|\varepsilon|+|c|)$, so the phase ends before $n_1$. Hence the whole phase lies within $O(|\varepsilon|+|c|)$ of the limit, and by Lemma 3.5 the order-one margins persist for $|\varepsilon| < \varepsilon_0$ and $|c| < c_0$. The margins of order $|c|$, (11) and (12), are signed and only increase during the phase. The front after the second triangle has up vectors $U_{\mathrm{a}}^+, U_{\mathrm{b}}^+$ and down vectors $D, U_{\mathrm{a}}^-, U_{\mathrm{b}}^-$. □

## 7. The second transition

This section collects the evidence for the delayed mechanism. Its target is the following statement, which we do not prove.

**Conjecture C** (Delayed pentagon)**.** *For every $\beta \in (\beta_{\min}, \pi/6)$ there are $\varepsilon_0 > 0$ and $K \geq 1$ such that for $0 < |\varepsilon| < \varepsilon_0$ and $0 < |c| < |\varepsilon|/K$, stacking from $F(\beta,\varepsilon)$ performs a pentagon transition after its front first has parastichy numbers $(3,2)$ and before it first has $(8,5)$, at most five transitions after the first triangle threshold of the $(3,2)$ phase.*

In four of the six first-order event calculations, and in the recorded simulations, the pentagon comes before $(3,5)$, so it interrupts the second Fibonacci transition. In the remaining two cases a tie that first-order analysis does not break decides whether it comes just before or just after the third triangle of step two; both branches are followed (Section 7.4). The calculation is not a certificate for the nonlinear dynamics: a candidate whose clearance vanishes at both retained orders can disappear at higher order, exposing a different lowest valid candidate. Section 7.4 ends with the remaining verification problem.

7.1. **What the first transition leaves behind.** Up/up and down/down ratios are frozen through every rhombic phase (Proposition 5.2), and each triangle multiplies its vector by $\omega$ or $\bar{\omega}$ up to $O(c)$, with corrections that agree to zeroth order for $U_{\mathrm{a}}$ and $U_{\mathrm{b}}$ (Lemma 3.3). Hence, at the end of the first transition,

$$\frac{U_{\mathrm{a}}^+}{U_{\mathrm{b}}^+} = e^{i\varepsilon}\big(1+O(c)\big), \qquad \frac{U_{\mathrm{a}}^-}{U_{\mathrm{b}}^-} = e^{i\varepsilon}\big(1+O(c)\big), \qquad \frac{D}{U_{\mathrm{b}}^-} = 1+O(c), \tag{13}$$

and by Proposition 5.2 again these ratios hold through the whole $(3,2)$ phase. In words: the two new up vectors still differ in angle by $\varepsilon$, and so do the two down vectors born from $U_{\mathrm{a}}$ and $U_{\mathrm{b}}$, while $D$ and $U_{\mathrm{b}}^-$ are parallel up to $O(c)$. In the sampled simulations, the up mismatch at the start of step two is well fitted by $\theta = |\varepsilon| - \kappa_1|c|$ with slope close to one and $\kappa_1$ between 1.399 and 1.666 for $\beta \in [0.42, 0.523]$ (Table 3); $\kappa_1$ collects the $O(c)$ terms of (13), and we have no closed form for it.

**Lemma 7.1** (The $(3,2)$ phase)**.** *Suppose no pentagon occurs during the $(3,2)$ phase. Then every step of the phase is the rhombus at the lowest lattice position, until a triangle candidate first becomes valid, at a notch hinge of $120^\circ + O(|c|)$; the front then lies within $O(|c|)$ of the front $K_\varepsilon(w)$ determined by (13), by one of the six cyclic words $w$ of the phase, by closure and by the threshold condition.*

*Proof.* By (13) the front is within $O(|\varepsilon|+|c|)$ of a $(3,2)$ rhombic lattice front with down vector $D$ and up vector $U$. Closure gives $3\operatorname{Im} D + 2\operatorname{Im} U = 0$, so lattice points $aU + bD$ have heights $\frac{1}{2}|\operatorname{Im} D|(3a-2b)$: distinct positions modulo the period are separated by at least $\frac{1}{2}|\operatorname{Im} D| \geq \frac{1}{2} b \sin 23.4^\circ > 0.19\, b$ along the whole phase, since $|\beta_D|$ decreases from about $41^\circ$ to $23.4^\circ$. Rhombus candidates are lattice positions, so the order of the passes is the lattice order with that margin. Triangle candidates overlap a neighbour while every hinge is below $120^\circ - K|c|$ for a constant $K$ (the exact switching boundary moves with $c$, as in Section 6). By assumption no pentagon is placed. The front thus follows the six-word cycle of the lattice, and at the first threshold crossing it is determined, up to $O(c)$, by the frozen ratios (13), the word, closure and the hinge $120^\circ$, up to $O(c)$. □

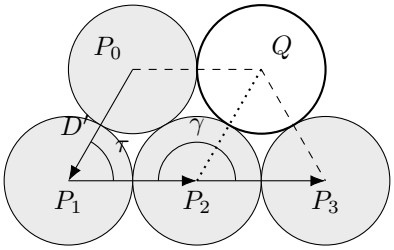



FIGURE 8. The degenerate pentagon. With hinges $\tau = 60°$ and $\gamma = 180°$ the disk $Q$ touching $P_0$ and $P_3$ also touches $P_2$ (dotted), and $|Q-P_1| = \sqrt{3}$. Turning $U_2$ up by $\theta$ lifts $Q$ off $P_2$; shrinking disks press it back by $\kappa_2|c|$.

The hypothesis isolates the unresolved case of Conjecture C: a pentagon in the $(3, 2)$ phase would already give its asserted breakdown. In the six words the two up vectors are never adjacent, so no one-down-two-up notch exists during the phase.

7.2. **When a pentagon fits.** Take a notch $P_0 \xrightarrow{D'} P_1 \xrightarrow{U_1} P_2 \xrightarrow{U_2} P_3$ with hinges $\tau = h(D', U_1) = 60° + \sigma$ (the cushion $\sigma$) and $\gamma = h(U_1, U_2) = 180° - \theta$ (the up mismatch $\theta$). At constant diameter a pentagon fits exactly when $\gamma + \tau \le 240°$ [5, Prop. 4.3], that is, when $\theta \ge \sigma$. The next lemma gives the first correction for shrinking disks.

**Lemma 7.2** (Pentagon with shrinking disks)**.** *Let $\beta_D$ be the bearing of $D'$. Near the degenerate point of Figure 8, the disk $Q$ touching $P_0$ and $P_3$ clears $P_2$ if and only if*

$$\theta \ge \sigma + \kappa_2(\beta_D)\,|c| + O\big(|c|(|\theta| + |\sigma| + |c|)\big), \qquad \kappa_2(\beta_D) = -\tfrac{1}{4}\big(\cos\beta_D + 3\sqrt{3}\sin\beta_D\big),$$

*and $|Q - P_1| = \sqrt{3}\,b_0$ at the degenerate reference configuration. Thus $P_1$ remains strictly clear in a sufficiently small neighbourhood.*

*Proof.* Work in the frame where $U_1$ points along the real axis, with unit diameter at $P_1 = 0$; heights are $\eta(x, y) = sx + ky$ with $s = \sin\alpha$, $k = \cos\alpha$, $\alpha = \arg U_1$ in the cylinder frame, and put $t = \sin(\alpha + \pi/3) = \frac{s}{2} + \frac{\sqrt{3}}{2}k$. At the degenerate point $P_0 = \omega$, $P_2 = 1$, $P_3 = 2$, $Q = \frac{3}{2} + \frac{\sqrt{3}}{2}i$. Let $G = |Q - P_2|^2 - \big(\frac{d_2+d_Q}{2}\big)^2$; $Q$ clears $P_2$ when $G \ge 0$.

*Along the mismatch* ($c = 0$): $P_3$ moves with velocity $i$ and $Q$ keeps touching $P_0$ and $P_3$. By (6) with zero right-hand sides, $Q' = (0, 1)$, so $\partial_\theta G = 2(Q - P_2) \cdot Q' = \sqrt{3}$.

*Along the rate* ($\theta = \sigma = 0$): the side lengths are mean diameters, which gives the velocities $P_2' = (\frac{s}{2}, 0)$, $P_3' = (2s, 0)$, $P_0' = \frac{t}{2}\,\omega$ (hinges held fixed). The tangencies with $P_0$ and $P_3$ give $Q_x' = \frac{s}{2} + \frac{5}{4}t$ and $\sqrt{3}\,Q_y' = \frac{6s+9t}{4}$, and then $\partial_c G = 2(Q - P_2) \cdot (Q' - P_2') - (\eta(P_2) + \eta(Q)) = \frac{5t-s}{2}$.

At $c = 0$ the boundary is exactly $\theta = \sigma$ [5, Prop. 4.3], so $G(\theta, \sigma, 0)$ vanishes on $\theta = \sigma$ and, $G$ being analytic, $G(\theta, \sigma, 0) = (\theta - \sigma)\big(\sqrt{3} + O(|\theta| + |\sigma|)\big)$. Adding the rate, $G = (\theta - \sigma)\big(\sqrt{3} + O(|\theta| + |\sigma|)\big) + c\big(\frac{5t-s}{2} + O(|\theta| + |\sigma| + |c|)\big)$, so the boundary $G = 0$ is $\theta = \sigma - \frac{5t-s}{2\sqrt{3}}\,c + O\big(|c|(|\theta| + |\sigma| + |c|)\big)$. With $\alpha = \beta_D + 2\pi/3$ one has $t = -\sin\beta_D$ and $\frac{5t-s}{2\sqrt{3}} = \kappa_2(\beta_D)$. Finally $|Q - P_1|^2 = \frac{9}{4} + \frac{3}{4} = 3$. □

A least-squares fit of the exact boundary over $\beta_D \in [-105°, -60°]$ returns the coefficients $-0.25000$ and $-1.29904$ of $\cos\beta_D$ and $\sin\beta_D$ (`verify/local_existence_fit.out`). At the steep child of the tiling $K_0$ below, $\cos\beta_D = 1/(2\sqrt{19})$, $\sin\beta_D = -5\sqrt{3}/(2\sqrt{19})$ and $\kappa_2 = 11/(2\sqrt{19}) \approx 1.262$.

7.3. **The tiling $K_0$ and the placement margins.** As $\varepsilon, c \to 0$ the burst of step two converges to the $(3, 2)$ threshold lattice with two triangles fired. The front at the moment of the pentagon is the word $U\,S\,U\,S\,U\,U\,D$ in the vectors (5), period $\sqrt{19}$, with points $z_0, \dots, z_7$ and $P_0, \dots, P_3 = z_3, \dots, z_6$ (Figure 9). In the rational model $Q = (11, 2)$, $T = z_6 + U = (18.5, 2.5)$ and $z_1 = (3.5, 1.5)$.

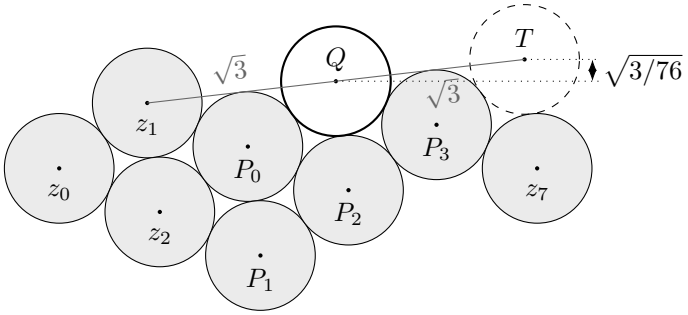
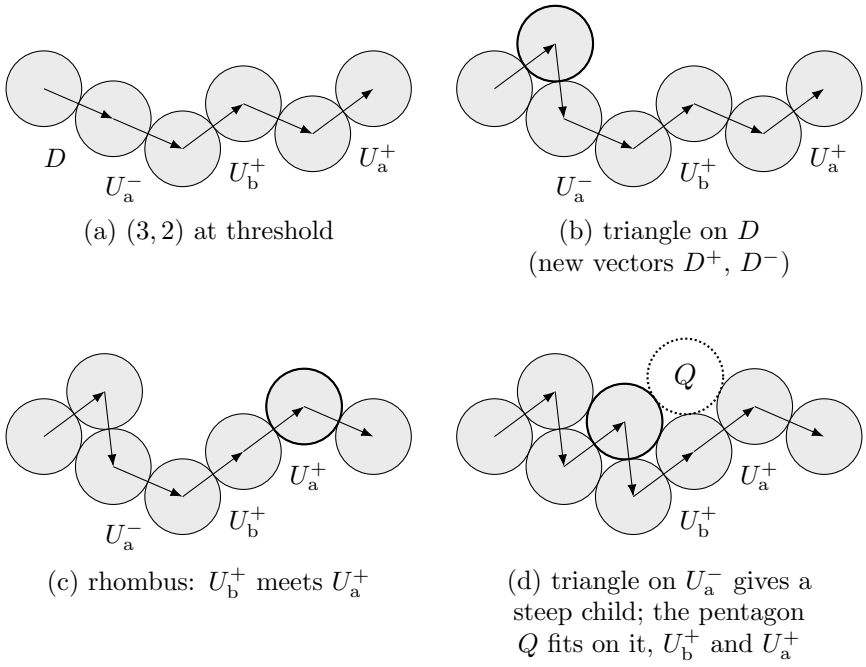



FIGURE 9. The tiling $K_0$ at the moment the pentagon is placed (unit diameters). The pentagon disk $Q$ touches $P_0$, $P_2$ and $P_3$. The lagging triangle $T$ on the last down vector is $\sqrt{3/76}$ higher, and the burst disk $z_1$ is at distance $\sqrt{3}$ and $\sqrt{3/76}$ lower.




(a) $(3,2)$ at threshold

(b) triangle on $D$ (new vectors $D^+$, $D^-$)

(c) rhombus: $U_{\mathrm{b}}^+$ meets $U_{\mathrm{a}}^+$

(d) triangle on $U_{\mathrm{a}}^-$ gives a steep child; the pentagon $Q$ fits on it, $U_{\mathrm{b}}^+$ and $U_{\mathrm{a}}^+$

FIGURE 10. The burst of step two at the $(3,2)$ threshold lattice (exact coordinates). A triangle on a down vector (b), a rhombus that brings the two old up vectors together (c), and a triangle whose steep child completes the notch on which the pentagon $Q$ fits (d). Panel (d) is the tiling $K_0$ of Figure 9.

**Lemma 7.3.** *In $K_0$, $Q$ touches $P_0$, $P_2$ and $P_3$ and is at distance $\sqrt{3}$ from $P_1$; $T$ touches $z_6$ and $z_7$; $|T-Q| = |Q-z_1| = \sqrt{3}$; and $\operatorname{Im}(T-Q) = \operatorname{Im}(Q-z_1) = \sqrt{3/76} \approx 0.1987$. Every other valid position is at least $2\sqrt{3/76}$ above $Q$.*

*Proof.* The first four statements are the rational identities $\Delta X^2 + 3\Delta Y^2 \in \{19, 57\}$ and $\Delta Y = \frac{1}{2}$. The last is an enumeration in exact rational arithmetic in the same model (`exact_limits.py`): the pentagon, rhombus and triangle at $Q$ are the same rational point, the three positions at $T$ likewise, and every other location has $\Delta Y \geq 1$, that is, lies at least $2\sqrt{3/76}$ above $Q$. □

7.4. **The burst.** It remains to follow the few events between the first threshold crossing of the $(3,2)$ phase and the pentagon. Figure 10 shows one of the six cases at the limit $\varepsilon = c = 0$.

For $|c| \ll |\varepsilon|$, a natural approximation starts at the constant-diameter threshold fronts $K_\varepsilon(w)$ of Lemma 7.1. The six words give six initial event calculations. Their positions are represented by

$$z(\varepsilon) = z_0 + \varepsilon z_1 + O(\varepsilon^2), \qquad z_0, z_1 \in \mathbb{Q}(\sqrt{3}, \sqrt{19})^2.$$

The lineage offsets in (13), closure, and the first notch at 120° determine the first-order input. A transverse apex is obtained by differentiating its two tangency equations. The accompanying `verify/burst_exact.py` computes these coefficients with rational field arithmetic and decides their signs by exact comparisons in the quadratic extensions. Floating-point approximations are not used for these sign or ordering decisions.

Under the event rules implemented in that script, the six words generate 34 leaves, each ending in a pentagon within at most five events and before $(8, 5)$. The audit records 60 non-parent clearances with zero constant and linear coefficients among selected candidates across the explored tree. The rules branch over equal-height candidates at both retained orders and select the triangle at an exact constant-diameter threshold. Two words admit a branch passing through $(3, 5)$ first. Runs at $\varepsilon = 10^{-5}$ and $10^{-7}$ with 60-digit arithmetic support these paths (`verify/burst_mp.py`); finite samples do not establish a uniform neighbourhood.

What is still needed. This calculation supports Conjecture C, but does not prove it. If a non-parent clearance has zero constant and linear coefficients, validity is unresolved at the retained orders. Treating the candidate as valid may hide a higher candidate that would become lowest if the former were excluded by a quadratic term. Branching over equal heights alone does not cover this possibility. Likewise, applying the constant-diameter threshold rule requires a separate check when the overshoot and the rate perturb it on the same scale.

A certificate for the conjecture must therefore track every unresolved clearance as well as every unresolved height comparison, prove that its event list covers all admissible outcomes, and propagate the incoming $O(|c|)$ uncertainty through those outcomes. Every terminating pentagon must have a positive leading clearance and a verified height comparison against all remaining candidates. An exact expansion to sufficient order, or interval bounds on a two-parameter neighbourhood, could provide such a certificate. The present first-order calculation is recorded as evidence, independently of the proved first-disk theorem.

## 8. The ratio law

Theorem B treats the first transition. The simulations below examine the second and suggest a narrow crossover in $r = \varepsilon/|c|$, consistent with the delayed mechanism of Conjecture C. They do not establish a sharp threshold.

For each rate $c$ and ratio $r = \varepsilon/|c|$ we ran thirty starting angles $\beta$, equally spaced in $[0.40, 0.52]$, and recorded the fraction $p(r)$ of them in which the second transition breaks (`verify/break_probability_consistent.py`). The starting angle stands in for where the discrete steps fall when a hinge crosses 120°, which the theory does not control; $p$ is a fraction of these chosen angles, not a probability for any natural measure. Figure 11 shows $p$ for three rates in steps of 0.1; for each rate, the sampled values satisfy $p = 0$ at $r \le 2.3$ and $p = 1$ at $r \ge 3.4$, and $p$ crosses $\frac{1}{2}$ between 2.85 and 3.1. Coarser runs at two more rates (`verify/break_probability_consistent_coarse.out`): at $c = -2 \times 10^{-3}$, $p(2.5) = 0$, $p(2.8) = 0.33$, $p(3.0) = 0.73$ and $p(3.5) = 1$; at $c = -1.25 \times 10^{-4}$, $p(2.5) = 0.17$, $p(2.8) = 0.83$ and $p = 1$ from $r = 3.0$. Across the sampled ratios at all five rates, $p = 0$ at $r \le 2.0$ and $p = 1$ at $r \ge 3.5$, and the midpoint wanders between about 2.65 and 3.1 with no monotone trend in $|c|$. We report the band as a description of these runs, not as a sharp constant: $K^*$ is about 3.

With the initial disks given a common diameter instead (the convention discussed after Definition 2.1) the band is $[2.4, 3.3]$ at the same rates. The convention matters for single runs near the edges: from $\beta = 0.523$, $\varepsilon = 0.005$, $c = -0.002$ the run with a common diameter completes the second transition and the run with diameters (2) breaks at $(3, 4)$.

A heuristic decomposition of the crossover follows the mechanism of Section 7. The pentagon fits when the inherited mismatch $\theta = |\varepsilon| - \kappa_1|c|$ exceeds the cushion $\sigma$ plus $\kappa_2|c|$ (Lemma 7.2), so the break happens near

$$\frac{|\varepsilon|}{|c|} \approx \kappa_1 + \kappa_2 + \frac{\sigma}{|c|}.$$

With $\kappa_1 \approx 1.40$ and $\kappa_2 \approx 1.26$ at $\beta = 0.523$, and $\sigma/|c|$ between about $-0.4$ and $0.7$ depending on the overshoot $\delta$ in (7), which is set by where the discrete steps fall when the hinge crosses

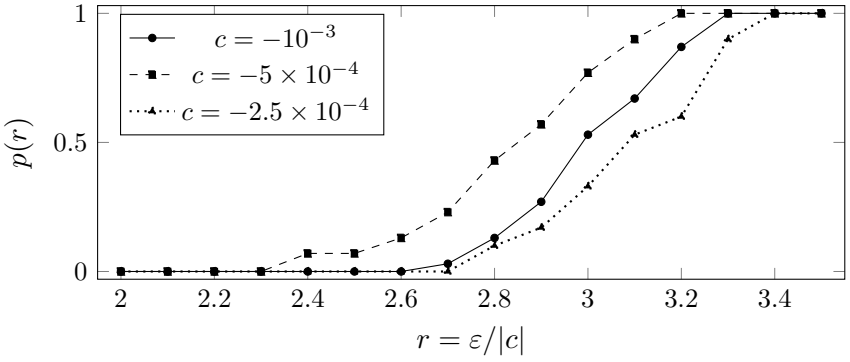



FIGURE 11. Fraction of thirty equally spaced starting angles $\beta \in [0.40, 0.52]$ for which the second Fibonacci transition breaks, against $r = \varepsilon/|c|$, every disk of diameter (2). For these sampled rates, angles and ratios, $p = 0$ at $r \leq 2.3$ and $p = 1$ at $r \geq 3.4$; values between grid points are not asserted.

120°, this gives 2.3 to 3.4, consistent with the band. The spread of the overshoot is what keeps $K^*$ from being sharp.

| $\beta$ | 0.42 | 0.45 | 0.48 | 0.50 | 0.523 |
|---|---|---|---|---|---|
| $\kappa_1 = (\varepsilon - \theta)/\lvert c\rvert$ | 1.665 | 1.622 | 1.568 | 1.518 | 1.399 |

TABLE 3. The $O(c)$ loss of mismatch through the first transition, every disk of diameter (2), stable to three digits over $c \in \{-2, -1, -0.5\} \times 10^{-3}$ and $\varepsilon/|c| \in \{5, 8\}$ (`verify/kappa1_consistent.py`). It depends on the seed convention: with a common initial diameter it runs from 1.584 to 1.657, increasing in $\beta$.

Two supplementary checks support the simulations, with limited scope. A second implementation, written independently (bisection for the implicit diameter, overlap tests against all disks of the last 400 steps), reproduces the same events step for step, including the pentagon at $(3, 4) \to (3, 3)$. Tolerance comparisons between 0 and $10^{-7}$ leave the tested event histories unchanged (`verify/independent_tolerance.out`); this is a finite reproducibility check. The 400-step overlap window is a truncation in that implementation, not a proved global bound. The lattice starts ($\varepsilon = 0$) generate their own irregularity of about 0.16 to 0.6 times $|c|$, below $K^*$, which is consistent with the Fibonacci transitions observed from regular fronts in the small-rate sweeps of [6].

## 9. Discussion

Theorem A rules out a uniform initial Fibonacci guarantee based on three independent upper bounds. It identifies the competing local scales and gives explicit fronts realising a first pentagon. It does not show that a mismatch-to-rate bound is necessary or sufficient for every front. The cancellation estimate explains persistence of a mismatch within a rhombic phase; it does not by itself determine which transition ends that phase. For the three-disk family, the numerical evidence motivates the following narrower statement.

> *Conjecture.* For every compact set of $\beta \in (\beta_{\min}, \pi/6)$ there are $K > 0$, $\varepsilon_0 > 0$ and $c_0 > 0$ such that for $0 < |c| < c_0$ and $|\varepsilon| < \min(\varepsilon_0, K|c|)$, stacking from $F(\beta, \varepsilon)$ completes the first two Fibonacci transitions.

Together with Conjecture C, if both were proved, this would identify $|\varepsilon|/|c|$ as the controlling parameter of the second transition, with the simulations putting the crossover between about 2 and 3.5. For general fronts we only ask: do fronts of parastichy numbers $(m, n)$ with $J \leq |c|/K$ and notch angles in a compact subset of $(60°, 120°)$ undergo $N$ Fibonacci transitions, for $K$ large

and $|c|$ small? Other open questions: a complete certificate for Conjecture C, including zero-clearance branches; whether breakdown always precedes $(3, 5)$, as the sampled runs indicate; a closed form for $\kappa_1$, the accumulated first-order residue of Proposition 5.2 through the first transition; whether a sharp threshold exists in a scaling limit that averages the overshoot; explicit $\varepsilon_0$ and $K$ for a proved delayed-breakdown statement; and the relation to the noise experiments of Swinton [10], where irregularity is renewed at every disk instead of being inherited once.

The biological interpretation is limited to this model. Its local geometry shows that slowing the diameter decrease reduces the clearance created at a transition. Relating this sensitivity to measured growth rates or sunflower defects would require a calibrated model and data beyond the computations reported here.

## Appendix A. Formal verification

The accompanying Lean 4 project specifies toolchain v4.34.0 and Mathlib. Its files verify the statements identified below, using the standard axioms (`propext`, `Classical.choice`, `Quot.sound`). Rechecking a file establishes its encoded statement; equivalence to the geometric model still requires the hypotheses and scope described here.

| Result | Lean file | main theorems |
|---|---|---|
| Lemma 2.2, $K_0$, Lemma 7.3 identities, $\sqrt{3/28}$ | `Tilings` | `threshold_tan`, `K0.height_gaps` |
| Lemmas 4.1, 4.4 (at $Q_0$), 4.5, 4.3 (notch); $\mu_Z > 0$ on valid fronts | `Static` | `notch60_tan`, `lattice_Q0`, `pentagon_first`, `notch_clear`, `channel_of_valid` |
| Local five-disk families: hypotheses satisfiable | `StaticExists` | `front_exists` |
| Lemma 3.1: differentiability at 0, tangency nearby and local uniqueness (not the full $C^1$ neighbourhood statement); non-vacuity | `Apex`, `Lemma4Exists` | `apex_exists`, `rhombic_family_exists` |
| Lemmas 3.2, 3.3, Corollary 3.4 | `FirstOrder` | `rhombic_step`, `triangle_step`, `cushion_rate` |
| Margin (11) and $\sqrt{7}$ | `FirstOrder`, `Lemma4` | `split_rate`, `pass_rate`, `n1_exclusion` |
| Lemma 7.2 | `Lemma4` | `dG_dc`, `dG_dθ`, `kappa2_eq`, `boundary_slope` |
| Lemma 5.1, gap in Lemma 6.2 | `PassBalance`, `Sweep` | `pass_balance`, `pass_lag`, `sweep`, `alternation_gap` |
| Proposition 5.2 | `Prop2` | `prop2` |

The local existence results in `Apex` and `StaticExists` do not certify periodic closure or global placement order. The following steps are proved by hand and remain outside Lean: the uniform remainders (Lemma 3.5); the full periodic first-disk argument, namely the ridge clearance (Lemma 4.3), the lattice order (Lemma 4.4), the separation of $Z$ and $E$, and the exclusion of all other positions (proof of Theorem A). Also outside Lean are the rational candidate enumerations; and the lattice-order argument in Lemma 7.1. The first-order burst exploration is also outside Lean and, as explained in Section 7.4, does not yet prove Conjecture C.

## Data and code availability

The simulators, recorded outputs, exact enumerations, first-order burst calculation and Lean source files are at [1], with a snapshot identifying the revision. The README there distinguishes reproduced checks from supplied records and lists the open proof obligations.

## Acknowledgements

Two AI systems were used in this work under the author's direction. Claude Opus 5.5 (Anthropic) developed the simulations, the first-order analysis and its proofs, the Lean formalisation and the draft. Codex (OpenAI, GPT-6) reviewed the draft, found an earlier first-disk pentagon on a five-disk front with one directional deviation zero, which prompted the construction of Theorem A on a different family of fronts, and contributed the revision that recast delayed breakdown as Conjecture C, computational checks and the identification of proof obligations. In version 2, a Codex referee report found the gap in the proof of Theorem B described below; Codex proved Lemma 6.1 and the strict bound for the words $U_{\mathrm{a}}\,U_{\mathrm{b}}^{-}\,D\,U_{\mathrm{b}}^{+}$ and $U_{\mathrm{a}}\,D\,U_{\mathrm{b}}^{-}\,U_{\mathrm{b}}^{+}$, including the treatment of the triple $D, U_{\mathrm{b}}^{-}, U_{\mathrm{b}}^{+}$, and Claude checked both against independent simulations and an independent derivation of the same comparison. The author is responsible for the content of this preprint, which has not yet been independently reviewed.

*Changes in version 2.* Version 1 asserted that the first triangle wins by an order-one margin and that the only near-tie in the $(2, 2)$ phase is the second triangle. Both are wrong: at each triangle event, and in two of the four words of the $(2, 2)$ phase, a triangle on a down vector coincides with the intended placement at the threshold tiling, and one two-down triple was not treated. Lemma 6.1 and the margin (12) close the gap, and Theorem B stands as stated. The triangle threshold is $120^\circ + O(|c|)$, not exactly $120^\circ$ (Section 6, Lemma 7.1); the first-disk theorem is stated for single-notch fronts $D^m U^n$; Swinton's version and sections are cited; the Lean coverage of Lemma 3.1 is stated exactly; the recorded outputs and the proof checks are completed and each exits nonzero on a failure (the explorations are records, not tests).

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